Eliciting Moral Preferences under Image Concerns: Theory and Experiment

Roland Bénabou   Armin Falk   Luca Henkel   Jean Tirole

March 23, 2026

Abstract

We analyze how the impact of image motives on behavior varies with two key features of the choice mechanism: single versus multiple decisions, and certainty versus uncertainty of consequences. Using direct elicitation (DE) versus multiple-price-list (MPL) or equivalently Becker-DeGroot-Marschak (BDM) schemes as exemplars, we develop a game-theoretic model to characterize how image-seeking inflates prosocial giving. The signaling bias (relative to true preferences) is shown to depend on the interaction between elicitation method and visibility level: it is greater under DE for low image concerns, and greater under MPL/BDM for high ones. We experimentally test the model's predictions and find the predicted crossing effect.

JEL codes: C91, D01, D62, D64, D78.

Keywords: Moral behavior, deontology, utilitarianism, consequentialism, social image, self-image, norms, preference elicitation, multiple price list, experiments.

Affiliations: Bénabou: Princeton University, NBER, CEPR, IZA, BREAD, and THRED. Falk: University of Bonn. Henkel: Erasmus University Rotterdam, CESifo, JILAEE, IZA, Tinbergen Institute. Tirole: Toulouse School of Economics (TSE) and Institute for Advanced Study in Toulouse (IAST).
Acknowledgments: We are grateful to Ingela Alger, Jean-François Bonnefon, Gary Charness, Franz Ostrizek, Pëllumb Reshidi, Marie-Claire Villeval and Joël van der Weele for valuable comments. Egshiglen Batbayar, Ana Luisa Dutra, Juliette Fournier, Mateo Rubio, Pierre-Luc Vautrey, Ben S. Young, and Youpeng Zhang provided superb research assistance.
Funding: Bénabou gratefully acknowledges financial support from the Canadian Institute for Advanced Study, Tirole and Falk from the European Research Council (European Community's Seventh Framework Program Grant Agreement no. 249429 and no. 340950, the European Union's Horizon 2020 research and innovation program, Grant Agreement no. 669217) as well as the German Research Foundation (DFG) through CRC TR 224 (Project A01).
Ethics approval: The study was approved by the ethical committees of the University of Bonn (no. 2019-01), Toulouse School of Economics and Princeton University (no. 11818).
Research transparency: The experimental instructions can be found at https://osf.io/354sb.

1 Introduction

Individuals' desire to signal that they are generous, caring, or generally "morally good," is a powerful driver of behavior. Accordingly, a large body of experimental research in both economics and psychology has examined how being observed by others affects prosocial behavior. While the *level* effect of observability on the prosociality of choices is well established (see, e.g., Bursztyn and Jensen, 2017, for an overview), little is known about how being observed *interacts* with the choice environment, in particular with respect to elicitation methods.

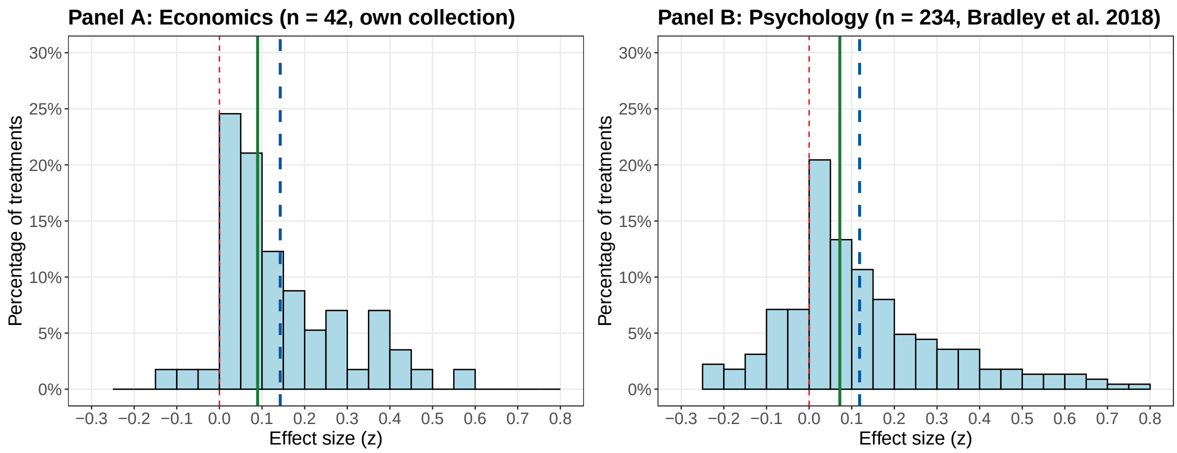
In the absence of image concerns, standard theory predicts that all incentive compatible elicitation schemes should yield identical results, revealing true preferences. Extending this intuition to experiments that study image concerns may be misleading, however, if different elicitation methods change the effective price of image. This conjecture motivates our study, as the presence of such an interaction can confound experimental findings and their implications. The same observability manipulation may thus generate large effects on a given prosocial behavior under one method and small ones under another, leading to opposite conclusions about the power of image. Such an image sensitivity of elicitation methods also has important practical ramifications for positive and normative interpretations based on answers to contingent-valuation surveys and for the design of visibility interventions.

To make progress on this question, we first develop a simple model to analyze how prosocial behavior can vary with the interplay between image concerns and elicitation methods. We then design an experiment to test the model's main predictions. Our theoretical and experimental analyses focus specifically on two widely used elicitation methods. The first, direct elicitation (*DE*), consists of a single choice implemented with certainty. The second is the Becker-DeGroot-Marschak method (*BDM*), for instance in its multiple-price list (*MPL*) format, which requires subjects to make multiple price-contingent decisions, one of which is randomly selected for implementation.

In the model, agents incur a cost to do good, or forfeit a "bribe" for causing harm. They also have image concerns, caring about appearing prosocial. Consequently, under either scheme, they engage in more prosocial behavior when the visibility of their choices increases. The novel contribution of the model is to bring to light three effects that make the two elicitation mechanisms differentially image sensitive and, when combined, generate a "crossing" pattern: when image concerns are low (but positive) *DE* will yield more contributions than *MPL*, and when they are high the ordering reverses. A secondary but notable implication is a greater image sensitivity (slope) of contributions under MPL than under DE. Relatedly, image-minded consequentialists will display deontological-like behavior –choosing the morally right action "at any (offered) price"– more readily under *MPL* than under *DE*.

To understand the effects at work, consider first a (DE-type) situation in which individuals may contribute to a cause (generate an externality $e > 0$) at some opportunity cost c,

Figure 1: Meta-analysis of image manipulation treatments on prosocial behavior



Notes: The figure displays the distribution of standardized effect sizes (Fisher's z) of observability treatment variations. Each observation corresponds to the reported treatment effect of increased observability on prosocial decision-making; higher values indicate a stronger positive effect on prosocial behavior. **Panel A** displays effect sizes from 42 treatment comparisons across 20 studies in economics. **Panel B** displays effect sizes from 234 treatment comparisons across 112 studies in psychology, taken from Bradley et al. (2018). Not displayed are 9 treatment comparisons that have effect sizes smaller than -0.3 or larger than 0.8. The dashed blue line represents the average across all treatments, the solid green line the median. For details on the meta-analysis, see Appendix Section A.

in time or money. In the relevant population there are two types, represented by Alice and Bob, who intrinsically value the cause at $v_H e$ and $v_L e < v_H e$. When social or self image concerns are present but not very strong, there is a range of prices $c > v_L e$ for which Bob will contribute in order to look as good as Alice, whereas for c' closer to $v_H e$ he will decline. In an *MPL/BDM* format, by contrast, the richer choice set and information thus generated make pooling more difficult, as Bob would have to state a willingness to pay of at least $v_H e$; for relatively low image concerns this is too high for him, so he will decline to contribute at *any* list price $c > v_L e$. This *discouragement effect* underlies the result that *MPL/BDM* yields less giving than *DE* when image concerns are positive but relatively weak.

Working in the other direction are two effects arising from the contingent nature of *MPL/BDM* bids, which effectively lower the purchase price of image. First, the randomly drawn list price $\tilde{c}$ could exceed one's bid c, making the latter partly *cheap talk*. This is related to random implementation, but more closely to the ability of participants in a public auction to "posture" with a high bid, while hoping that someone else will outbid them. Second is what we term the *cheap-act* effect: conditional on a bid c being binding ex-post, the average price paid is only $E[\tilde{c}|\tilde{c} \leq c]$. As image concerns intensify, Bob's desire to pool and Alice's desire to separate lead to increasingly high bids, so the cheap-talk effect weakens (implementation becomes more certain). In contrast, the cheap-act effect strengthens (for standard distributions the "discount" $c - E[\tilde{c}|\tilde{c} \leq c]$ grows), causing *MPL* contributions to rise above those under *DE*. Hence, a crossing effect.

On the experimental side, image effects have been studied using a wide array of designs, including elicitation methods that rely on single versus multiple choices and on deterministic versus stochastic implementation. This heterogeneity is associated with considerable variation in estimated effects, as illustrated in Figure 1. Panel A presents a histogram from a meta-analysis we conducted focusing on economic experiments. Each observation corresponds to the normalized treatment effect of increased observability on prosocial behavior, covering 42 treatment comparisons from 20 papers. Panel B reports results from Bradley et al. (2018), who analyze 234 treatment comparisons drawn from 112 experiments primarily from psychology. Across both sets of studies, two key patterns emerge. First, observability generally increases prosocial behavior. Second, effect sizes vary substantially, raising the question of whether part of this heterogeneity could be driven by differences in elicitation methods. In this respect, the evidence in Figure 1 is only suggestive, because the studies differ not only in elicitation methods but also in the prosocial behaviors studied, the image manipulations, and other important design features. A clean test of the model's predictions requires an experimental setup that holds all other aspects of the choice task constant, while systematically varying the elicitation method. This is what we do in our experiment.

In the experiment, about 700 participants face a choice between: (i) directing a $350€$ donation to a charity in India that will use the money to treat five tuberculosis patients, resulting statistically in the expected saving of one human life; or (ii) taking money for themselves, where the amount is either a fixed $100€$ under *DE*, or determined by the subjects' cutoff on an *MPL* where prices range from 0 to $200€$. These two elicitation conditions are crossed with low and high moral-image treatments. Specifically, in the *Low Image* conditions, subjects choices are double-blind (i.e., cannot be linked to their identity, not even during the payment procedure), so that no social image, only self image, is operating. In contrast, in the *High Image* conditions, subjects know prior to their decision(s) that their decisions will be observed and judged by other people.

Comparing the fractions of subjects choosing the "saving a life" contribution over taking $100€$, we find a sizeable reversal between *DE* and *MPL* as image concerns go from weak to strong, as predicted by the theory. In the *Low Image* treatments, the fraction opting to save a life is 48% under MPL versus 59% under DE, while in the *High Image* condition it is 63% under DE versus 72% under *MPL*. On the cautionary side, statistical significance is only at the 6-7 percent level. Concerning the broader prediction of differential image sensitivity, we find a clear effect, in the expected direction: as observability increases, the rise in contributions is significantly more pronounced under *MPL* than under *DE* ($p < 0.01$). In any case, our simple experiment should be seen as proof-of-concept for the mechanisms brought to light by the model, opening them up to more systematic exploration. We also conduct a robustness experiment with 366 additional subjects where we keep all aspects of the decision environments unchanged, except that choices are now over a non-moral good (a university voucher), for which no image concerns arise. As expected, we find no significant difference between the two elicitation methods, showing that our results are specific to image-relevant decisions.

1.1 Related Literature

Previous research on social and self image has primarily focused on how they spur prosocial behaviors, and how this signaling incentive is affected by the presence of rewards (Bénabou and Tirole, 2006, 2011, 2025; Ariely et al., 2009; Ashraf et al., 2014; Galeotti et al., 2021; Falk, 2021) or excuses (Dana et al., 2007; Exley, 2016; DellaVigna et al., 2012; Grossman and van der Weele, 2017; Bursztyn et al., 2018; Garcia et al., 2020). Our analysis highlights instead their interaction with the mechanism through which choices are made. Not only are schemes such as *DE* vs *MPL/BDM* differentially sensitive to image concerns, but their effectiveness at measuring intrinsic preferences, or on the contrary spurring higher contributions, can even reverse as reputational motives intensify.

Another strand of work focuses on decision makers' probability of being pivotal (Feddersen et al., 2009; Grossman, 2015; Falk et al., 2020; Bartling et al., 2024), which relates to what we term the cheap-talk effect. We show how, in mechanisms such as *MPL*, the probability of having one's choice implemented varies systematically with the intensity of image concerns, as does the expected cost at which the choice will be implemented, and we analyze how both effects shape equilibrium behavior. This relates the paper to work on auctions with signaling, in which bidders seek to demonstrate goodness, wealth, or a strong aftermarket position (Goeree, 2003; Giovannoni and Makris, 2014; Bos and Pollrich, 2025; Bos and Truyts, 2023). In our setting, an agents' distribution of potential outcomes depends only on his own choices, and this lower strategic complexity allows us to identify intuitive effects and testable predictions.

With respect to experimental methodology, we contribute to the study of alternative elicitation mechanisms. Substantial research has compared how *DE*, *BDM*, *MPL* or random implementation (Selten, 1967) affects behavior in one-shot, anonymous games such as dictator or public-goods (Brandts and Charness, 2011; Chen and Schonger, 2016).[1] There is also a large body of research on elicitation methods for risk, time and ambiguity preferences (Charness et al., 2013; Cox et al., 2015; Cohen et al., 2020; Baillon et al., 2022). To our knowledge, no such study has explored these methodological issues with reputationally sensitive decisions like those analyzed here.[2] For choices in the moral domain, self-image (at

[1]Concerning DE with deterministic versus random implementation (an intermediate case relative to MPL), the overview by Charness et al. (2016) reports generally ambiguous effects. As the model will make clear, it is only in the presence of sufficient signaling concerns that probabilistic implementation will matter. In contrast, risk attitudes play no role in the effects that we identify, which directly affect expected returns.

[2]Closest to this is Grossman and van der Weele (2017), with a result that relates to what we term the discouragement effect. When subjects in a "moral wiggle room" game (Dana et al., 2007) must state contingent choices for each of the two payoff structures that might occur, they are more willing to learn ex post which one obtained than when making this choice ex ante in the original game, because ignorance of a tradeoff can no longer serve as an excuse to preserve (self) image.

least) is almost inevitably at play, and can create differences between elicitation methods.[3]

Finally, the paper relates to the debate between consequentialist and deontological principles. The evidence on how people behave in practice is mixed: the literature on public-goods contributions and charitable giving finds that choices are generally sensitive to the implied consequences (Ledyard, 1995; Goeree et al., 2002), including the risk of having no impact (Brock et al., 2013), overhead costs (Gneezy et al., 2014) and the link to utility (Chakraborty and Henkel, 2024). At the same time, there is evidence of "warm glow" altruism, in which utility is derived from the act as such (Andreoni, 1989, 1990). Experiments that directly focus on consequentialist versus deontological or expressive choices (Van Leeuwen and Alger, 2024; Chen and Schonger, 2022; Falk et al., 2020; Bénabou et al., 2026) also suggest a mix of preferences. Our paper shows that, when image concerns are important, a mechanism like *MPL* or *BDM* can easily lead consequentialist agents to adopt deontological-looking behaviors.

2 Model

We study how people's (un)willingness to accept different tradeoffs between personal gain and harm to others varies across two "canonical" types of preference-elicitation mechanisms: direct elicitation and BDM/multiple-price list. We look for Perfect Bayesian Equilibria of the corresponding games, and in case of multiplicity: (i) apply the D1 criterion; (ii) when that is insufficient, select the Pareto-dominant equilibrium.[4] All proofs are gathered in the Appendix.

2.1 Direct Elicitation

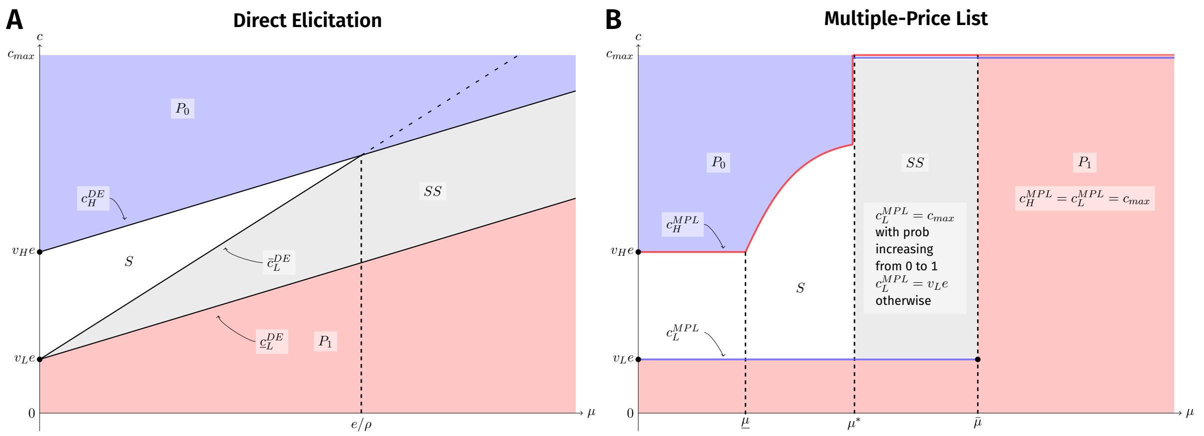
We first consider decisions under DE, introducing agents' basic preferences in the process. Agents are risk-neutral, with a two-period horizon, $t = 1, 2$. At date 1, an individual faces a take-it-or-leave-it opportunity to engage in prosocial behavior $(a = 1)$ or act selfishly $(a = 0)$. Choosing $a = 1$ involves a personal cost $c > 0$ but generates a public good or externality $e \geq 0$. Agents differ in their intrinsic motivation to act morally: given e, it is either $v_H e$ (high type) or $v_L e$ (low type), with probabilities ρ and $1 - \rho$, $v_H > v_L \geq 0$, and average $\bar{v} = \rho v_H + (1 - \rho) v_L$.[5] Besides the externality, the second feature of action

[3]In the non-moral domain, in contrast, the literature tends to find no difference between *DE* and *BDM* (Miller et al., 2011; Berry et al., 2020; Cole et al., 2020).

[4]While Brandts and Holt (1992) and Banks et al. (1994) find that refinements such as the intuitive criterion and D1 are unreliable to predict subjects' behavior in certain signaling games, those where this happens are constructed to violate the single-crossing property (they feature regions of "reverse type dominance"). Our *DE* and *MPL* games have the single-crossing property, and for such games both papers find that refinements work rather well. The games in these papers also have only two actions and all Nash equilibria involve pooling, whereas under *MPL* the high type can raise their bid to the least-cost separating level.

[5]Appendix (Section 6.2) discusses how the paper's mechanisms and results translate in richer type spaces.

Figure 2: Equilibrium under Direct Elicitation (panel A) and Multiple-Price List (panel B)



Notes: P_0: pooling at $a_H = a_L = 0$; S: separation, $a_H = 1, a_L = 0$; SS: semi-separation, $a_H = 1, a_L \in (0,1)$; P_1: pooling at $a_H = a_L = 1$.

$a = 1$ tying it to the moral domain is that it can be reputationally valuable, conferring a social or self-image benefit at date 2. In the social context, the agent knows his type but the audience (peer group, firms, potential partners) does not. In the self-signaling context, he has an immediate, "intuitive" sense of his deep preferences at the moment of action – for instance, how much empathy or spite he experiences – but later on the intensity of that feeling is imperfectly accessible ("forgotten"), and only the deed itself, $a = 0$ or 1, can be reliably recalled to assess his own moral identity.

Under either interpretation, an agent of type $v = v_H, v_L$ has expected utility

$$(ve - c)\, a + \mu \hat{v}(a), \tag{1}$$

where: (i) $\hat{v}(a)$ is the expected type conditional on the action $a \in \{0, 1\}$ and the cost faced; (ii) μ is the strength of self or social-image concerns, common to all agents, reflecting both the visibility of their actions and how much they care about appearing prosocial. This utility may be additively augmented by any externalities generated by others, but since that term is independent of the agent's action we omit it here. Note that these preferences are consequentialist: an agent's desire to behave prosocially trades off the impact he expects his actions to have, the personal cost involved, and the reputational consequences.

When an action is off the equilibrium path, as can occur in a pooling equilibrium, we will apply forward induction, in the form of the D1 refinement.As common in signaling models, multiple equilibria may still coexist: thus, when

$$\max\{v_L e - c + \mu(v_H - v_L),\ v_H e - c + \mu(v_H - \bar{v})\} \le 0 \le v_H e - c + \mu(v_H - v_L),$$

there is both a pooling equilibrium at $a = 0$ and a separating one in which the v_H type contributes, with a mixed-strategy one in-between, all robust to D1. When such multiplicity persists but the equilibria are Pareto-comparable, we choose the one that is best for both types, namely the no-contribution pooling equilibrium. Indeed, separation yields lower payoffs for both, since $\mu v_L < \mu \bar{v}$ and $v_H e - c + \mu v_H \leq \mu \bar{v}$. The same two-step selection process (first D1, then Pareto among D1 survivors) will be applied throughout the model's analysis.

As illustrated in Panel A of Figure 2 (for $\rho < 1/2$), equilibrium behavior is then characterized by three cost (or incentive) thresholds, increasing in the reputational concern μ, that delineate regions of separation, semi-separation, and pooling:

$$v_H e - c_H^{DE}(\mu) + \mu\,(v_H - \bar{v}) \equiv 0, \tag{2}$$

$$v_L e - \bar{c}_L^{DE}(\mu) + \mu\,(v_H - v_L) \equiv 0, \tag{3}$$

$$v_L e - \underline{c}_L^{DE}(\mu) + \mu\,(\bar{v} - v_L) \equiv 0. \tag{4}$$

Denoting $a_H^{DE}(c,\mu)$ and $a_L^{DE}(c,\mu)$, or a_H and a_L for short, the two types' probabilities of choosing $a = 1$, we show

Proposition 1. *The outcome of direct elicitation is as follows:*

1. *For low costs, $c < \min\{\underline{c}_L^{DE}, c_H^{DE}\}$, everyone behaves morally, $a_H = a_L = 1$.*
2. *For intermediate costs, $c \in (\underline{c}_L^{DE}, c_H^{DE})$, if any, the high type behaves morally $(a_H = 1)$, but the low type's probability $a_L(c)$ of doing so decreases with c, and then equals 0 for $c \geq \min\{\bar{c}_L^{DE}, c_H^{DE}\}$.*
3. *For high costs, $c \geq c_H^{DE}$, both types behave immorally, $a_H = a_L = 0$.*

Relative to "pure" (intrinsic) moral preferences ve, decision thresholds are inflated due to reputational concerns; see (2)-(4). In particular, the range of costs $[\bar{c}_L^{DE}, c_H^{DE}]$ where full separation occurs shrinks with μ, becoming empty for $\mu > e/\rho$.

2.2 Multiple-Price List

Under *BDM*, the individual "names his price" by stating what maximum cost $c \in [0, c_{\max}]$ he is willing to incur for taking action $a = 1$, where $0 \leq v_L e < v_H e < c_{\max}$. Equivalently, c represents his willingness to accept a "bribe" to make the immoral choice, $a = 0$. This elicitation is made incentive-compatible by drawing some $\tilde{c} \in [0, c_{\max}]$ according to a pre-announced distribution $G(\tilde{c})$, and implementing $a = 1$ at cost $\tilde{c}$ only when $\tilde{c} \leq c$. With *MPL*, subjects state contingent choices at each possible level of $\tilde{c}$. Both schemes generate identical incentives, so we gather them under the label of *MPL*, since that is the format we implement experimentally.[6] In experiments, the distribution G is typically uniform, but here we allow any other case, including $c_{\max} = +\infty$.

[6]We implement *MPL* imposing a single switching point, which is strategically equivalent to *BDM*.

The objective function of an agent with preference $v \in \{v_L, v_H\}$ now takes the form:

$$\mathbb{E}_G\left[(ve - \tilde{c})\mathbb{1}_{\{\tilde{c} \le c\}}\right] + \mu\mathbb{E}[v|c]. \tag{5}$$

Let $L(c)$ denote the low type's loss (not inclusive of image) from selecting a cutoff c greater than than his true valuation $v_L e$:

$$L(c) \equiv \int_{v_L e}^{c} (\tilde{c} - v_L e)\, dG(\tilde{c}) = \underbrace{\mathbb{P}(\tilde{c} \in [v_L e, c])}_{\text{cheap-talk effect}} \underbrace{(\mathbb{E}(\tilde{c}|\tilde{c} \in [v_L e, c]) - v_L e)}_{\text{cheap-act effect}} \tag{6}$$

and assume $L(c_{\max}) < \infty$, for which it suffices that $E_G[\tilde{c}] < \infty$. The first term is the *cheap-talk effect*: the randomly drawn price $\tilde{c}$ could exceed the bid c, in which case no contribution occurs and no loss is incurred. The second term is the *cheap-act effect*: when $\tilde{c} < c$, the agent must contribute, but the cost is only $E[\tilde{c}|\tilde{c} \le c] < c$ on average, reducing the loss relative to a *DE* contribution at price c. We will say that a subject is *observationally deontological* if he turns down all prices on the proposed list (with distribution G): given the available data, he behaves as someone who would not act immorally "at any price."

We now solve for both types' willingness to accept (WTA) under the multiple-price list, denoted c_H^{MPL} and c_L^{MPL} respectively. Note first that, *absent* reputation concerns ($\mu = 0$), MPL and DE are equivalent, and reveal true preferences: $c_H^{DE} = c_H^{MPL} = v_H e,\ \underline{c}_L^{DE} = \bar{c}_L^{DE} = c_L^{MPL} = v_L e$. For $\mu > 0$, comparing $L(c)$ to $\mu(v_H - v_L)$ and $\mu(v_H - \bar{v})$, namely the reputation gap in a separating equilibrium and the high type's gain if they were to separate from the low type in a pooling equilibrium, yields a unique (D1 refined) equibrium. The two types' corresponding strategies are illustrated in Panel B of Figure 2, and characterized again by critical thresholds between separating, semi-separating and pooling regions:

$$\underline{\mu} \equiv \frac{L(v_H e)}{v_H - v_L} < \mu^* \equiv \frac{L(c_{\max})}{v_H - v_L} < \frac{L(c_{\max})}{\rho(v_H - v_L)} \equiv \bar{\mu}. \tag{7}$$

Proposition 2. *The outcome of the MPL mechanism is as follows:*[7]

1. *When the reputational concern μ is low, $\mu < \mu^*$, the high type's WTA for behaving immorally is $c_H^{MPL} = \max\{v_H e,\ L^{-1}(\mu(v_H - v_L))\}$, while the low type finds it too costly to pool and accepts $c_L^{MPL} = v_L e$.*

   Initially, for $\mu \le \underline{\mu}$, separation is costless for the high type, then as μ rises he has to raise his reservation price to separate from the low type.

2. *When μ is intermediate, $\mu \in [\mu^*, \bar{\mu}]$, the high type can no longer separate and becomes observationally deontological, $c_H^{MPL} = c_{\max}$. The low type randomizes, with probability*

[7] Under *DE*, c is an exogenous price, whereas under *MPL* it is an equilibrium bid. Thus, whereas Proposition 1 most naturally varies the offer c, Proposition 2 varies the parameter μ. If desired, Panel A of Figure 2 shows how to recast Proposition 1 in terms of μ: for all μ there exist $\underline{c}(\mu) \le c^*(\mu) \le \bar{c}(\mu)$ such that P_1 obtains for $c \le \underline{c}(\mu)$, SS in $(\underline{c}(\mu), c^*(\mu)]$, S in $(c^*(\mu), \bar{c}(\mu)]$, and P_0 for $c > \bar{c}(\mu)$.

$a_L(\mu)$ increasing in μ, between that same "virtuousness" ($c_L^{MPL} = c_{\max}$) and revealing himself (accepting $c_L^{MPL} = v_L e$).

3. *When $\mu > \bar{\mu}$, (self) image concerns are strong enough that both types' behavior is observationally deontological: $c_H^{MPL} = c_L^{MPL} = c_{\max}$.*

2.3 Comparison of DE vs. MPL

Under both elicitation schemes, image concerns naturally raise contributions, as seen in Figure 2. More novel are the following questions:

1. Is one elicitation scheme *more image-sensitive* than the other?
2. Which one yields *more expected contributions?*

Formally, at a given cost $c \in [0, c_{\max}]$, what fraction of people $\bar{a}^{DE}(c, \mu)$ accept forfeiting c to implement $a = 1$ under *DE,* versus what fraction $\bar{a}^{MPL}(c, \mu)$ state a willingness to pay of at least c under *MPL*? And how does $\bar{a}^{DE}(c, \mu) - \bar{a}^{MPL}(c, \mu)$ depend on μ?

While the answers generally depend on the specific value of c, the cases of sufficiently low and high image concerns yield clear predictions. We will denote as μ^{**} the solution to $\underline{c}_L^{DE}(\mu) = c_{\max}$, or

$$\mu^{**} \equiv \frac{c_{\max} - v_L e}{\bar{v} - v_L} > \frac{L(c_{\max})}{\bar{v} - v_L} = \bar{\mu}. \tag{8}$$

Putting together the results of Propositions 1 and 2, we have:

Proposition 3. *For each type $\tau = H, L$,*

1. *Visibility raises contributions: for any $c \in [0, c_{\max}]$, $a_\tau^{DE}(c, \mu)$ and $a_\tau^{MPL}(c, \mu)$ coincide at $\mu = 0$, then both increase (weakly) as μ rises, reaching 1 for μ large enough.*

2. *Under low image concerns, DE yields more contributions: for all $\mu \in (0, \underline{\mu})$, $a_\tau^{DE}(c, \mu) \geq a_\tau^{MPL}(c, \mu)$, with strict inequality for $c \in (v_L e, \bar{c}_L^{DE}(\mu))$ and $c \in (v_H e, c_H^{DE}(\mu))$, both nonempty.*

3. *Under high image concerns, MPL yields more contributions: for all $\mu \geq \bar{\mu}$, $a_\tau^{DE}(c, \mu) \leq a_\tau^{MPL}(c, \mu) = 1$, with strict inequality for $\tau = L$ and $c \in (\underline{c}_L^{DE}(\mu), c_{\max})$, which is nonempty whenever $\mu \in (\bar{\mu}, \mu^{**})$.*

4. *The average behavior over types, $\bar{a}^m(c, \mu) \equiv \rho a_H^m(c, \mu) + (1 - \rho) a_L^m(c, \mu)$, $m = DE$, MPL, inherits these same properties.*

The first result is standard, while the others stem from the interplay of three effects.

Weak image concerns: discouragement effect dominates. When $\mu > 0$ is low enough that separation under *MPL* is costless, we have $c_H^{MPL}(\mu) = v_H e < c_H^{DE}(\mu)$ and $c_L^{MPL}(\mu) = v_L e < \underline{c}_L^{DE}(\mu)$, hence the second result. Intuitively, *MPL* raises the cost to the low type of mimicking the high one to at least $v_H e$, and for low reputational concerns (small μ) such a discrete cost is not worth it. Under *DE*, in contrast, matching the high types' decision to contribute at low prices c yields small but proportionate reputational gains. This intuition is reflected in the fact that the lower boundary of the separating region is linear in Panel A of Figure 2, whereas it is initially flat in Panel B.

Strong image concerns: cheap-act effect dominates. At high values of μ, reputational concerns become paramount, and the cost of signaling is lower under *MPL* than under *DE,* since high values of c must only be paid with a probability less than 1: the effective cost of stating a cutoff c is only $E\left[\tilde{c}|\tilde{c} \leq c\right] < c$. It is even bounded by $L(c_{\max}) + v_L e < \infty$, which limits the extent to which the high type can separate, so that for $\mu > \bar{\mu}$ full pooling occurs: $c_H^{MPL} = c_L^{MPL} = c_{\max}$, so $a^{MPL}(c,\mu) = 1$, whereas $\bar{a}_L^{DE}(c,\mu) < 1$ as long as $\mu < \mu^{**}$. Most importantly:

Property 1. *For any distribution satisfying the monotone hazard rate property* $(g/(1-G)$ *increasing), the "discount"* $c - E\left[\tilde{c}|\tilde{c} \leq c\right]$ *is increasing in* c. *Therefore, as* μ *rises and with it each type's cutoff, the* cheap-act *effect becomes stronger, which increases MPL contributions relative to DE.*

Intermediate image concerns. Inside $(\underline{\mu}, \bar{\mu})$, a third *"cheap-talk"* effect is also important. Under *MPL,* an agent who states a cutoff $c < c_{\max}$ has only a probability $G(c) < 1$ of being called upon to actually "deliver": if $\tilde{c} > c$ is drawn, he neither incurs a cost *nor* generates the externality e. This makes it safer to state high cutoffs, thus adding to the *cheap-act* effect. The latter is not as strong in this range as for high values of μ, and conversely the *cheap-talk* effect weakens as μ rises, pushing $G(c^{MPL})$ closer to 1. The net balance of the three effects is generally ambiguous in this intermediate range, and consequently so is the sign of $a^{DE} - a^{MPL}$.

Implications. Three main predictions emerge from the model. First, as usual, greater visibility increases contributions. Second, at low but positive levels of visibility, *DE* leads to more prosocial outcomes, as the *discouragement effect* dominates. Third, at high levels (but not so high as to push everyone to $a = 1$ under *DE*), this ordering reverses: *MPL* induces more moral decisions, due to the now dominating *cheap-act* effect.

The inequalities in Proposition 3 can be weak or strong, depending on the region of the parameter space. This is a standard feature of models with discrete types and action spaces, which typically disappears when there is sufficient heterogeneity to span all cases. For this reason, when confronting the model with data, we will tighten the predicted inequalities to be strict ones.[8]

[8]Our tests will vary $(M,\mu) \in \{DE,\ MPL\} \times \{\mu_L, \mu_H\}$ while maintaining the same realized cost c. By Proposition 3, the set of parameters such that $0 < \mu_L < \underline{\mu} < \bar{\mu} < \mu_H < \mu^{**}$ and $\bar{a}^{DE}(c,\mu_L) - \bar{a}^{MPL}(c,\mu_L) > 0 > \bar{a}^{DE}(c,\mu_H) - \bar{a}^{MPL}(c,\mu_H)$ is nonempty provided that $\underline{c}_L^{DE}(\bar{\mu}) < c_H^{DE}(\underline{\mu})$, which reduces to $L(c_{\max}) - (1-\rho)L(v_H e) < (v_H - v_L)\, e$. With a uniform G, for example, a sufficient condition for any ρ is that $c_{\max}/e \in \left(v_H - \left(v_H^2 - v_L^2\right)^{1/2}, v_H + \left(v_H^2 - v_L^2\right)^{1/2}\right)$.

3 Experimental Design

We designed an experiment to test the model's main predictions, in particular whether sensitivity to image concerns varies with the mode of elicitation and the strength of image concerns as described in Proposition 3. For this purpose, we need a design that confronts subjects with a significant moral decision, varying both the elicitation procedure and the importance of social image.

3.1 Saving a Life

To generate situations in which one choice is unambiguously perceived as the more moral one, we adopt the *Saving a Life* paradigm from Falk and Graeber (2020). In the paradigm, subjects can either take money for themselves or implement a fixed, life-saving donation to a charity dedicated to the treatment of tuberculosis in India. According to the World Health Organization, tuberculosis is one of the ten leading causes of death worldwide, even though there are highly effective antibiotic treatments available. Together with the Indian non-profit organization *Operation ASHA*, we calculated a specific monetary amount sufficient to identify, treat, and cure a number of patients such that – in expectation – one patient will be saved from death by tuberculosis due to the donation. Combining public information on the charity's operations with estimates from peer-reviewed studies on mortality due to tuberculosis and treatment effectiveness for the specific type of treatment and location considered (Straetemans et al., 2011; Tiemersma et al., 2011; Kolappan et al., 2008), we determined that level to be 350€: by allowing for the treatment of five patients, such a donation allows the (expected) saving of one human life. Without the donation, conversely, five patients are not treated who would otherwise have been, implying that in expectation one will succumb to the disease.

This paradigm thus contrasts the option of saving a life (major positive externality e) by triggering a donation of 350€ ($a = 1$) versus that of taking money for oneself at a opportunity cost c ($a = 0$), inducing a clear tradeoff between morality and self-interest. This is reinforced by the donation being cost-effective: the amount is well above all monetary payments possible for the subjects themselves, as described later, and the money is directly used to treat patients, without any administrative or transaction cost.

Furthermore, the potentially extreme consequences of the donation decision also increase the likelihood that subjects will take the choice task and its signaling/reputational implications seriously. To further ensure that subjects make choices with sufficient deliberation, we provided them with comprehensive explanations about the disease, the treatment, the veracity of the experiment, and so on. In particular, the experimental instructions explain in detail the consequences of the donation and, conversely, what the absence of the donation entails (for details, see the Instructions in Appendix Section E). As our experimental results will show (Section 4.2), subjects do take this information and the decision seriously. If they were not convinced of the credibility or effectiveness of the donation, they would simply take the money for themselves, no matter how little was offered. However, as indicated by choices in the *MPL* treatments, subjects show a clear sensitivity to the monetary amount offered, and even at 200€ under anonymity, there are still a substantial fraction of subjects who do not take the money. For related findings, see Falk and Graeber (2020).

3.2 Treatments

To test the model's predictions, we use a 2×2 between-subjects design, varying the elicitation method (DE vs. MPL) as well as the visibility and moral salience of choices (*Low Image* vs. *High Image*) at the payment stage.

Under direct elicitation (DE), subjects faced the binary choice between receiving $c = 100$€ (≈\$110) as payment, or saving a human life in expectation. As part of the experimental design, we predetermined this single value of $c = 100$€ as a compromise between two practical concerns: (i) c must be high enough to generate choices of both types; (ii) in contrast to *MPL*, each implemented decision has a sure cost to the experimental budget of either c or the full 350€ donation, which quickly adds up.

For the MPL conditions, we used a price-list design: starting with $c =$ 0€ and proceeding in 10€ increments up to $c =$ 200€, subjects could indicate in each of the 21 contingent choices whether they wanted to save a life or take c for themselves. Each price was then equally likely to be drawn for implementation (uniform G).[9] Figures D.1 and D.2 in Supplementary Appendix D display the corresponding decision screens.

Turning to visibility, recall that the two key forces underlying Proposition 3, namely the *discouragement* and the *cheap-act* effects, both require a non-zero level of image concerns. To ensure a minimal level of image concern in both treatments, we notified subjects at the start that: (i) they were anonymously paired with another participant in the same session; (ii) they would see, at the end of the experiment, their own and their partner's choices displayed alongside on their screens, as would their partner. Apart from observing the partner's choices, subjects received no information about them, so that no other aspect of the dyad would influence decisions.

Social-image image can still arise, however, in experiments where the final payment procedure is not anonymous. In such cases, the experimenter can at least partly infer choices from the amounts paid, and link them to subjects' identity when interacting with them; due to the binary nature of the decisions in our experiment, choices are in fact perfectly revealed through the payments. To keep image concerns minimal in the *Low Image* treatment $(\mu = \mu_L)$, we made the payment procedure double-blind, so that not even the experimenter

[9] To avoid inconsistent answers, we enforced a single-switching rule: if a subject chose to save a life for a given monetary value, the computer filled in the same choice for all lower values. However, participants could always revise their decisions after the autocompletion, and had to confirm their final choices before moving on.

could link subjects' decisions to their identity. Following Barmettler et al. (2012), at the start of each session one subject was randomly designated to carry out all payments: they did not participate in the regular experiment, and thus had no knowledge about the choices. At the end, payments were stuffed into envelopes and the selected subject handed them out, in an adjacent room, to those who had actively participated.

The *High Image* treatment ($\mu = \mu_H$), in contrast, was designed to induce strong image concerns. Prior to making their decisions, subjects were informed that upon receiving payment: (i) their choice would be observed and compared to that of their matched partner by a committee consisting of three persons, sitting in the room where payments would take place; (ii) both partners' choices would be projected onto a wall, and they would have to read both aloud, using two predetermined sentences. To fix their beliefs and provide a justification for the existence of the committee, we explained to subjects that the committee was there to rate the prosociality of their decisions.

3.3 Procedure

In total, 697 subjects (405 female, mean age = 24.01, SD = 6.21) participated in 36 sessions at the BonnEconLab of the University of Bonn: 178 in the *MPL-Low Image* treatment, 178 in *MPL-High Image*, 165 in *DE-Low Image*, and 176 in *DE-High Image*[10]. Subjects were recruited using Hroot (Bock et al., 2014), and the experiment was conducted using oTree (Chen et al., 2016). Sessions lasted about 60 minutes, with a show-up fee of 12€. For each session, one matched pair of subjects was randomly drawn, and their choices implemented. Thus, in the DE treatments, each of the two either received 100€, or triggered a life-saving 350€ donation. In the *MPL* treatments, one price from the list was randomly drawn (uniformly), and the pre-stated choices of both partners for this price were implemented. Therefore, each one either triggered the donation or received up to 200€.[11]

At the beginning of each session, subjects received a brief verbal introduction to the experiment. In the *Low Image* treatments, the procedure ensuring anonymity was explained and demonstrated. In the *High Image* treatments, the committee setup was shown. Subsequently, all subjects received detailed information about tuberculosis, its effects, and treatment. The instructions also referred them to a website where they were invited to confirm the validity of the information. We then introduced the charity and its working procedure, and explained our calculations regarding the life-saving effect of the 350€ donation. Subjects then learned about their choice options and, after answering a few comprehension

[10]At 80% power ($1 - \beta = 0.80$) and a significance level of $\alpha = 0.05$, this sample size of just under 180 subjects per treatment implies that we are powered to detect effect sizes of approximately $d = 0.30$ (Cohen's d).

[11]This random implementation adds another layer of the cheap-talk effect, but one that affects *DE* and *MPL* in exactly the same way (formally equivalent to dividing μ by the probability of implementation), and thus leaves all comparisons between the two unaffected. We further discuss this feature and its potential effects on choices in Section 4.3.

questions, made their decisions. Finally, they completed a short questionnaire and were paid in a separate room, with payment procedures depending on treatment status, as explained above. For further details on the procedure and instructions, see Supplementary Appendix E.

4 Hypotheses and Results

Our outcome variable is the fraction $\bar{a}^m(c, \mu)$ of subjects who choose to save a life over receiving c, given an elicitation method $m \in \{$*DE, MPL*$\}$ and a level of visibility $\mu \in \{\mu_L, \mu_H\}$. For brevity, we will refer to $\bar{a}^m(c, \mu)$ as "total contributions".

4.1 Hypotheses

Based on Proposition 3, we state:[12]

Hypothesis 1. *For both* DE *and* MPL, *total contributions are higher under* High Image *than under* Low Image: $\bar{a}^{DE}(c, \mu_H) > \bar{a}^{DE}(c, \mu_L)$, $\bar{a}^{MPL}(c, \mu_H) > \bar{a}^{MPL}(c, \mu_L)$.

Hypothesis 2. *Under* Low Image, *total contributions are higher under* DE *than under* MPL*:* $\bar{a}^{DE}(c, \mu_L) > \bar{a}^{MPL}(c, \mu_L)$.

Hypothesis 3. *Under* High Image, *total contributions are higher under* MPL *than under* DE*:* $\bar{a}^{DE}(c, \mu_H) < \bar{a}^{MPL}(c, \mu_H)$.

Hypothesis 1 captures the standard effect of signaling concerns. The novel ones are Hypotheses 2 and 3, reflecting the dominance of the *discouragement effect* at μ_L and the *cheap-act effect* at μ_H. Together, they constitute the model's distinctive crossing prediction, which we will test at $c = 100$€, as explained earlier.

4.2 Results

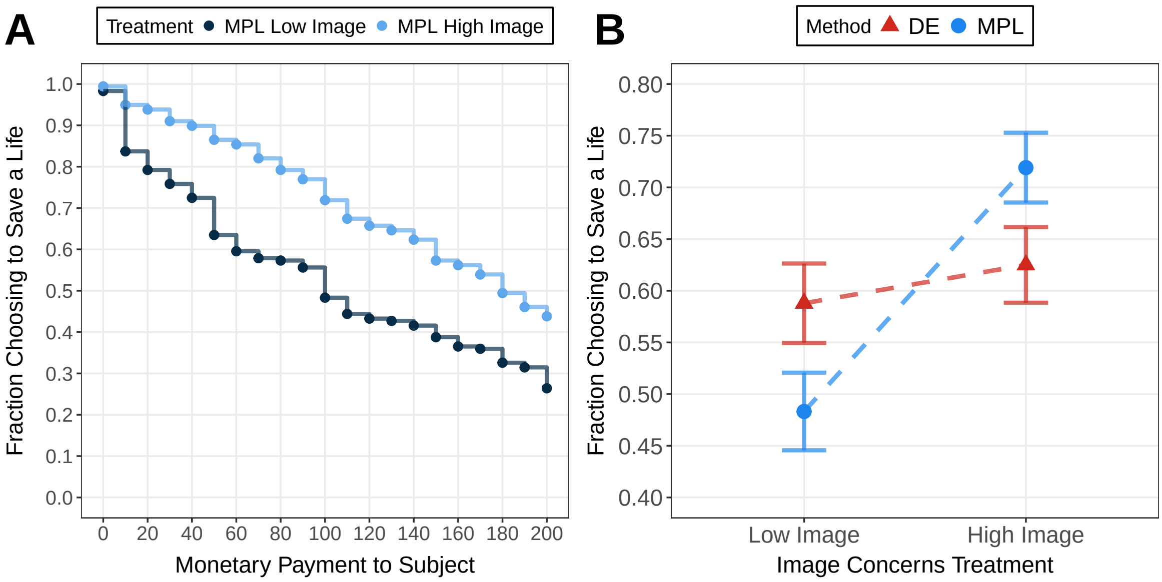
Hypothesis 1. Under both elicitation methods, increased visibility led to a rise in total contributions, but the magnitude was markedly different. Under *DE*, 58.8% of subjects chose to save a life in *Low Image* and 62.5% in *High Image* – a relatively small and insignificant increase ($p = 0.51$, Fisher's exact test[13]).[14] Under *MPL*, increased visibility had a much

[12]Recall from Section 2 that these predictions rely on our equilibrium selection, which uses D1 and Pareto dominance. The appropriateness of the first criterion for our games was discussed in footnote 4, and the latter seems rather intuitive.

[13]We follow the convention of reporting two-sided tests, even though we are testing the directed (inequality) hypotheses emanating from the model. Unless otherwise noted, we use Fisher's exact test.

[14]This corresponds to a standardized effect size of $z = 0.038$ (Fisher's z). Interestingly, this effect is comparable to the average effect size observed in similar studies included in our meta-analysis (Appendix Section A). Among the 12 treatment comparisons that also used *DE*, the average effect size is $z = 0.028$.

Figure 3: Main Experimental Results



Notes: **Panel A** displays the fractions of subjects that choose to save a life for each offered price in the *MPL Low Image* and *MPL High Image* treatments. **Panel B** shows the interaction effect of elicitation method and image concerns. Displayed is the fraction of subjects who choose to save a life instead of receiving 100€ for themselves either when this choice is the only choice they make (*DE*), or when this choice is embedded in a price list (*MPL*), and depending on whether the choice is made in the *Low Image* or the *High Image* treatment. Error bars indicate the standard error of the mean.

larger effect. At almost all payment levels, the fraction of subjects choosing to save a life is at least 15 pp. higher under *MPL-High Image* than under *MPL-Low Image*, resulting in significantly different distributions ($p < 0.001$, Kolmogorov–Smirnov test); see Panel A of Figure 3. At 100 €, contributions are 23.6 pp. and significantly higher under *High Image* than under *Low Image* ($p < 0.001$).

Hypotheses 2 and 3. Panel B of Figure 3 shows that the fractions $\bar{a}^m(100, \mu)$ choosing to save a life over 100€ clearly differ by elicitation method, with the ranking reversing between μ_L and high μ_H. Under *Low Image*, we observe $\bar{a}^{MPL}(\mu_L) < \bar{a}^{DE}(\mu_L)$, as predicted by Hypothesis 2, and consistent with the dominance of the *discouragement effect*. The difference is large, with the fraction saving a life rising from 48.3% to 58.8% between *MPL* and *DE*, though significance is slightly below the conventional level ($p = 0.065$, Fisher's exact test). Conversely, under *High Image* we observe $\bar{a}^{MPL}(\mu_H) > \bar{a}^{DE}(\mu_H)$, in line with the *cheap-act effect* dominating, as predicted by Hypothesis 3. The difference is again about 10 percentage points, but now in the opposite direction, rising from 62.5% under DE to 71.9% under MPL, albeit again with significance slightly short of 5% ($p = 0.070$).

Table 1, Panel A regresses the probability of choosing to save a life (instead of taking 100€) on a dummy for the type of elicitation (1 for *MPL*), which yields a positive coefficient for *Low Image* in Column (1), and a negative one for *High Image* in Column (3).[15] Columns (2) and (4) show that these effects remain largely unaffected by controls for age, gen-

[15]The results remain qualitatively unchanged with Probit or Logit regressions.

der, high-school graduation grade, highest educational degree obtained so far, self-reported monthly income, and a measure of religiousness (Likert scale).

Hypotheses 2-3 represent the strictest possible test of the model – a particular ordering of four variables – which may explain the marginal significance of these results. A more standard test concerns their joint implication of a *differential image sensitivity*: as image rises from μ_L to μ_H, the increase in contributions should be more pronounced for *MPL* than for *DE*. Panel B of Table 1 thus presents an OLS regression interacting *High Image* with *MPL*, using *DE-Low Image* as baseline; the interaction is positive and significant at the 1-percent level.

Overall, the results lend support to the key predictions of the model emanating from Proposition 3, albeit with significance being sometimes marginal. As such, our simple experiment can be taken as a proof of concept for the novel mechanisms brought to light by the model, thereby opening them up to further exploration.

4.3 Robustness Experiment

One may worry that features of the elicitation methods unrelated to image concerns influence our results. Note first that such features would have to generate not just different *DE* versus *MPL* contributions, but also a flipping of that gap as image rises from low to high, which seems unlikely. Thus, features of elicitation methods can at most confound one of our comparisons, not the crossing effect itself. The previous literature has identified three factors that could potentially confound the comparison between the two elicitation methods.

First, in our experiment, only a subset of subjects had their decision implemented for real. In the MPL treatments, another randomization takes place, which is absent in DE: if selected for payout, one decision of the price list is randomly selected. If subjects violate the independence axiom and view these two randomization processes not separately but rather as a meta-lottery, this could potentially affect the comparison. This issue is also present in the many experiments that study decisions over lotteries and pay only one lottery out for real. In this context, it is usually assumed that subjects evaluate the different random processes in isolation, an assumption that has been repeatedly validated empirically[16]. It is natural to assume that subjects also perceive the two processes in isolation in our experiment, since they were introduced and explained at two different points in the instructions.

The second factor is the so-called compromise effect (Andersen et al., 2006; Birnbaum, 1992; Simonson, 1989). When presenting a price list, the focus lies perceptually on the center. This in turn could change the attractiveness of the options appearing in the middle of the price list, biasing answers away from the subject's true valuations. To control for this effect, we carefully selected the DE value to correspond to the value precisely in the

[16] See e.g., Starmer and Sugden (1991), Cubitt et al. (1998), and Hey and Lee (2005).

Table 1: Regression analyses of the effect of the elicitation method on prosocial behavior

Panel A:				
Dependent variable:	Choice to Save a Life (vs. 100€)			
	Low Image		*High Image*	
	(1)	(2)	(3)	(4)
MPL	-0.105^{*}	-0.103^{*}	0.094^{*}	0.091^{*}
	(0.054)	(0.053)	(0.050)	(0.050)
Constant (*DE*)	0.588^{***}	0.626^{***}	0.625^{***}	0.622^{***}
	(0.038)	(0.049)	(0.037)	(0.046)
Controls		X		X
Observations	343	343	354	354

Panel B:		
Dependent variable:	Choice to Save a Life (vs. 100€)	
	(1)	(2)
MPL	-0.105^{*}	-0.097^{*}
	(0.054)	(0.053)
High Image	0.037	0.052
	(0.053)	(0.052)
MPL X High Image	0.199^{***}	0.190^{***}
	(0.073)	(0.072)
Constant (*DE Low Image*)	0.588^{***}	0.595^{***}
	(0.038)	(0.044)
Controls		X
Observations	697	697

Panel C:		
Dependent variable:	Choice of Voucher (vs. 10€)	
	(1)	(2)
MPL No-Image	0.045	0.051
	(0.047)	(0.047)
Constant	0.253^{***}	0.227^{***}
	(0.033)	(0.047)
Controls		X
Observations	366	366

Notes: The table shows OLS regression coefficients. The dependent variable in Panel A is an indicator variable equal to one if the subject chose a donation that saves a human life and zero if the subject chose 100€ for themselves. "MPL" is an indicator variable equal to one if the subject was part of the *MPL* treatment and zero if the subject was part of the *DE* treatment. Columns (1) and (2) display the results for the *Low Image* treatment, and columns (3) and (4) for the *High Image* treatment. Panel B adds the variable "High Image", which is an indicator variable equal to one if the subject was part of the *High Image* treatment and zero if the subject was part of the *Low Image* treatment. The dependent variable in Panel C is an indicator variable equal to one if the subject chose a voucher for a university online shop and zero if the subject chose 10€ for themselves. "MPL No-Image" is an indicator variable equal to one if the subject was part of the *MPL No-Image* treatment and zero if the subject was part of the *DE No-Image* treatment. Robust standard errors in parentheses. Controls include age, gender, income, religiousness, educational level, and high school grade.

middle of the price list in the MPL treatments. As such, it seems unlikely that differences in perceptions could explain discrepancies between the elicitation methods.

Third, subjects' comprehension could vary between elicitation methods, potentially leading to differences in behavior. For instance, it is well-known that people behave differently under alternative complex auction mechanisms that are strategically equivalent (Kagel et al., 1987; Li et al., 2017). For this reason, instead of the Becker-DeGroot-Marschak method, which shares features of a second-price sealed-bid auction, we use a price list, which is easier to understand (Andersen et al., 2006). Here, research has shown that comprehension is high, even in field settings (e.g., Burchardi et al., 2021). Their results support the notion that MPL is not substantially less understood by subjects compared to DE. Our experiment also includes comprehension questions, allowing us to assess the results' sensitivity to participant confusion. Using incorrect answers as a proxy for confusion, we restrict the sample to subjects who made at most one mistake, which eliminates 21 of them. Rerunning our analyses reveals that, if anything, the effects become slightly stronger. In particular, the difference between *MPL* and *DE* becomes significant at the 5% level for both the *Low Image* and *High Image* cases. See Appendix Table B.1 for details.

To summarize, we would not expect differences between DE and MPL in our experiment once image concerns are absent. In order to document this empirically, we conducted a robustness experiment, which is explained next.

Design. For the robustness experiment, we used a good that is unrelated to prosocial and moral considerations, so that image concerns are plausibly absent. For this non-moral good, we chose a 35 € voucher for the University of Bonn's online shop. With the voucher, subjects can buy sweatshirts, T-shirts, and accessories related to the university. The voucher cannot be returned and is only valid for purchases in the shop. There were two between-subject treatments: *DE No-Image* and *MPL No-Image*. In the former, subjects could choose between 10 € and the voucher, while in the latter they faced a price list from 0€ to 20€ in 1€ increments. Note that this closely mimics the decisions in the main experiment. The only difference is that all values are divided by 10. As in the main experiment, subjects were paired with another subject and were given each other's decisions at the end of the experiment.

Accordingly, instructions for the decisions were identical, with the sole difference being that descriptions related to the saving a life paradigm were replaced with descriptions of the voucher. Consequently, any factors influencing the comparison between *DE* and *MPL* in the main experiment should also manifest in the robustness experiment.

Procedure. Subjects were recruited from the same subject pool as the main experiment, with the restriction that they had not previously participated in the main experiment. The experiment was conducted as a virtual lab experiment since in-person lab sessions were not possible due to the ongoing Covid-19 pandemic. That is, the experiment started and ended at a pre-specified date and time, and the experimenter was available during the experiment in case of problems.

In total, 366 subjects (227 female, mean age 26.88, SD 7.87) took part, 188 in the *MPL No-Image*, and 178 in the *DE No-Image* treatment, respectively. The experiment lasted on average 13 minutes, for which the subjects received a show-up fee of 3€. Subjects were grouped in virtual sessions consisting of roughly 24 subjects, and one pair was randomly selected for payout out of each virtual session. Exactly as in the main experiment, for these two subjects, either their DE decision was implemented or a randomly chosen decision from the MPL.

Results. Assessing subjects' general valuation of the voucher, we observe considerable variation in switching behavior in the *MPL No-Image* treatment. In total, 76% had an interior switching value, meaning they preferred the voucher in the initial decision but switched to preferring the monetary value at some point. This is very similar to behavior in the *MPL-Low Image* treatment, where 72% of subjects had an interior switching point. Comparing the choice at 10 € in *MPL No-Image* with *DE No-Image*, we find that 29.8% choose the voucher in *MPL* and 25.3% in *DE*. This difference is small in magnitude and not statistically significant ($p = 0.35$; two-sided Fisher's exact test). It is also in the opposite direction of what we find in the main experiment for the *Low Image* case, which is the natural comparison. Comparing effect sizes (Cohen's d), the difference between *MPL Low Image* and *DE Low Image* has an effect size of $d = 0.21$, while the effect size is $d = -0.10$ for the difference between *MPL No-Image* and *DE No-Image*. Accordingly, not only is the effect in the opposite direction, but also has a small effect size. Panel C of Table 1 replicates this null result in an OLS-regression, with column (2) using the same variables as control variables as in the main experiment, compare Table 1, columns (2) and (4). Thus, we do not observe any meaningful differences between the two elicitation methods in our setting once image concerns are removed.

5 Conclusion

Our model and experiment show that image concerns affect the measurement of moral preferences in ways that interact with the elicitation method. Regardless of whether one is interested in image-inclusive preferences (for positive predictions) or in purely intrinsic ones (for normative judgments), these results have potentially important implications.

On the one hand, suppose the primary objective is to identify "true" moral preferences from choice data. In this case, the decision environment needs to be designed to minimize the scope for image concerns. While anonymity or even double-blind protocols can largely eliminate social-image and experimenter-demand effects, self-image preservation often remains an important motive for subjects engaged in moral decisions (see Bénabou and Henkel, 2025, for a survey). When there is such a presumption, our results point toward

MPL rather than *DE*.[17]

On the other hand, suppose the primary objective is to maximize the behavioral effect of image concerns on prosocial behavior. This may apply to researchers whose design requires a strong and robust image effect as a first stage. To illustrate, Hofmeier and Strang (2023) study the role of spillovers of a visibility intervention on follow-up behavior. Without an initially strong treatment effect of the intervention, no spillover effects can be analyzed. In such cases, combining high visibility with an *MPL* may facilitate the objective.[18] This consideration is also relevant for practitioners. Consider a charity aiming to maximize donations, or a government seeking to increase contributions to public goods by altering the observability of choices. Rather than asking individuals to make a publicly visible donation of a fixed amount, mechanisms such as a MPL or an auction, which elicit self-chosen contributions under uncertainty, have the potential to generate higher total contributions.

An interesting question for future research is whether the patterns identified in this paper extend to other types of moral decisions. Here, we focused on people's willingness to engage in tradeoffs between their private costs or benefits $(c$ and $\mu\hat{v})$ from some action and its social consequences (e). Another important type of moral tradeoff concerns "trolley-like" problems which ask, e.g., whether it is permissible to sacrifice one life to save a larger number (Foot, 1967; Thompson, 1976). More generally, choices are between alternative public goods $(e$ vs. $e')$, and departing from some default action causes harm to some people but can be a means to achieve a greater good, $(e' > e)$. Trolley-like problems have received considerable attention. Large international surveys have explored multiple variants, with the aim of informing the kinds of tradeoffs that may be faced by autonomous vehicles, robots, and other AI systems (e.g., Awad et al., 2018). Recently, real-stakes versions of the trolley and similar consequentialism-versus-deontologism dilemmas have been experimentally studied (Bénabou et al., 2026).

It remains an open question, however, how image concerns affect choices for this kind of moral tradeoff. Does being observed increase or decrease individuals' willingness to accept harm to others in order to achieve greater good? And how do such image effects depend on the decision-environment, in particular the elicitation method? To illustrate, consider one of the ends-versus-means tradeoffs studied in Bénabou et al. (2026): Using a *DE*-method, the authors ask subjects to make a repugnant statement (in private) in order to trigger a €15 donation to a children's cancer charity. 44 percent of subjects choose the "deontological" option, refusing to make the repugnant statement. It is unclear how these responses would

[17]Prior methodological discussions have largely emphasized a practical tradeoff: *MPL* yields richer and potentially more precise choice information, whereas *DE* is simpler to implement and explain. Our results introduce an additional consideration, namely the interaction between self-image and elicitation procedure.

[18]Regardless of the researcher's specific objective, our results underscore the importance of using elicitation methods consistently when estimating treatment effects. Whenever image concerns cannot be ruled out, the standard prediction that elicitation methods yield identical results no longer holds. Hence, if elicitation methods are not kept constant across treatments, estimated treatment effects may be confounded by the interaction between image concerns and the elicitation method itself.

change if subjects instead faced a price menu in an *MPL*-setting, with donation amounts, e.g., ranging from €1 to €15 in €1 increments.

More generally, if researchers seek to obtain quantitative estimates of prosocial or moral preferences, the choice of elicitation method may be less innocuous than previously assumed.

6 Appendix

6.1 Proofs

Proof of Proposition 1. From (2)-(4), it follows that:

(P_0) : $a_H = a_L = 0$, sustained by out-of equilibrium belief (OEB) $\hat{v} = v_H$ following $a = 1$ (by the D1 criterion), is an equilibrium if and only if $c \geq c_H^{DE}$. When

$$\bar{c}_L^{DE} = v_L e + \mu(v_H - v_L) \leq c \leq v_H e + \mu(v_H - v_L) \equiv \bar{c}_H^{DE},$$

it coexists with a separating equilibrium S in which $a_H = 1 = 1 - a_L$, plus a mixed-strategy one in-between. A shown earlier, however, P_0 is Pareto dominant, and therefore selected.

(P_1) : $a_H = a_L = 1$, sustained by OEB $\hat{v} = v_L$ following $a = 0$ (by D1), is an equilibrium if and only if $c \leq \underline{c}_L^{DE}$.

(S) : $a_H = 1 - a_L = 1$ is an equilibrium if and only if $\bar{c}_L^{DE} \leq c \leq \bar{c}_H^{DE}$.

(SS_1) : $0 < a_L < 1 = a_H$, with belief $\hat{v} \in (v_L, \bar{v})$ following $a = 1$, is an equilibrium if and only if $\underline{c}_L^{DE} < c < \bar{c}_L^{DE}$. The low type's mixed strategy $a_L(c) \in (0, 1)$ is then given by combining the indifference condition $v_L e - c + \mu(\hat{v}(a_L) - v_L) = 0$ and the Bayesian posterior $\hat{v}(c) = [\rho v_H + (1-\rho) a_L v_L] \,/\, [\rho v + (1-\rho) a_L]$:

$$v_L e - c + \frac{\mu\rho(v_H - v_L)}{\rho + (1-\rho) a_L(c)} \equiv 0, \tag{9}$$

so $a_L(c)$ decreases with c, while the reputation $\hat{v}(c)$ following $a = 1$ increases.

(SS_0) : $0 = a_L < a_H < 1$, with beliefs $\hat{v} \in (\bar{v}, v_H)$ following $a = 0$, is an equilibrium if and only if $c_H^{DE} < c < \bar{c}_H^{DE}$. It always coexists with P_0, and is always dominated by it.

These results jointly imply that:

(a) If $\underline{c}_L^{DE} < \bar{c}_L^{DE} < c_H^{DE}$, the unique equilibrium is P_1 for $c < \underline{c}_L^{DE}$; SS_1 for $c \in [\underline{c}_L^{DE}, \bar{c}_L^{DE}]$; and S for $c \in [\bar{c}_L^{DE} < c_H^{DE}]$. For $c \geq c_H^{DE}$, the dominant equilibrium is P_0.

(b) If $\underline{c}_L^{DE} < c_H^{DE} < \bar{c}_L^{DE}$, the unique equilibrium is P_1 for $c < \underline{c}_L^{DE}$, and SS_1 for $c \in [\underline{c}_L^{DE}, c_H^{DE}]$. For $c > c_H^{DE}$, the dominant equilibrium is P_0.

(b) If $c_H^{DE} < \underline{c}_L^{DE} < \bar{c}_L^{DE}$, the unique equilibrium is P_1 for $c < c_H^{DE}$, and for $c \geq c_H^{DE}$ the dominant equilibrium is P_0. ■

Proof of Proposition 2. The proof of existence is standard. For example, for a separating equilibrium to obtain, it must be: that (i) type v_L obtains his symmetric-information allocation (otherwise, he would be better off selecting $c_L^{MPL} = v_L e$), and (ii) he does not want to mimic type v_H: $\mu(v_H - v_L) \leq L(c_H^{MPL})$ and $c_H^{MP} < c_{max}$. It is easily verified that the proposed strategies satisfy these conditions, and similarly for the semi-separating and pooling equilibria .

The equilibrium is not unique absent refinement, however. For example, there is a pooling equilibrium at $c^{MPL} = v_H e < c_{max}$ when $\mu(\bar{v} - v_L) \geq L(v_H e)$, sustained by OBE $\hat{v} = v_L$ following any declared price $c \neq v_L e$. Note, however, that sorting implies monotonicity, so there is at most one price, denoted c^*, that can be chosen with positive probability by both types; any other price claimed by type v_H (respectively, v_L) exceeds c^* (respectively, lies below it) c^*) . Denote $\hat{v}(c)$ the mean belief following a price c, and consider a deviation to $c' = c^* + \varepsilon$, for $\varepsilon > 0$ arbitrarily small, together with the set of belief responses that raise both types' utilities relative to equilibrium

$$\hat{V}_L \equiv \{\hat{v}(c^* + \varepsilon) \mid \mu\left[\hat{v}(c^* + \varepsilon) - \hat{v}(c^*)\right] > L_L(c^* + \varepsilon) - L_L(c^* + \varepsilon)\},$$
$$\hat{V}_H \equiv \{\hat{v}(c^* + \varepsilon) \mid \mu\left[\hat{v}(c^* + \varepsilon) - \hat{v}(c^*)\right] > L_H(c^* + \varepsilon) - L_H(c^* + \varepsilon)\}.$$

Clearly $V_L \subset V_H$, so by D1 the deviation must induce a probability-one belief on v_H; thus, the only possible pooling price is $c = c_{max}$. Consequently, the equilibrium must take one of the three forms described in the proposition, and because it is obtained on disjoint sets of parameters, it is unique under D1. ■

6.2 Richer type spaces

Our two-type model brings to light three channels through which image and choice mechanisms interact. With more types they still operate, though less can be said in general about their net balance when comparing *DE* and *BDM*. The cheap-talk and cheap-act effects arising under *MPL*, one attenuating and the other strengthening with image concerns, are both very general, extending even to a continuum: equilibrium bids naturally rise with μ, which increases the implementation probability and reduces the effective price of image; see (6). For the discouragement effect, with $n > 2$ types it remains the case that, for μ positive but low enough, *MPL*'s richer information hinders pooling. With a continuum, however, separation can no longer be costless, for any reputational stakes. In Supplementary Appendix C we show that, with a uniform distribution over $[0, v_{max}]$:

1. The characterization of *DE* (Proposition 1) carries over, with type v now contributing at c if $b^{DE}(v) \equiv v + \mu(E[v'|v' > v] - E[v'|v' < v]) > c$, defining a threshold $v^*(c, \mu)$ decreasing in μ.

2. So does that of *MPL* (Proposition 2), except for the interval of costless revelation. As

with discrete types, equilibrium involves: (i) separation up to some $v^{\dagger}(\mu)$, decreasing in μ, with bids solving $b^{MPL}(0) = 0$ and $b^{MPL}(v) = \arg\max_b\{-\int_v^b(\tilde{c} - v)g(\tilde{c})d\tilde{c} + \mu\hat{v}(b)\}$, hence $b'(v)[b(v) - v] = \mu/g(b(v)) > 0$; (ii) observationally deontological pooling at $b^{MPL}(v) = c_{max}$ by all $v > v^{\dagger}(\mu)$.

3. In Proposition 3, the first and third results are unchanged: contributions under both schemes are sincere for $\mu = 0$, then increase continuously with μ, for each type and at any cost level (H1); and *MPL* delivers more contributions than *DE* for large μ (H3), as the cheap-act effect induces observationally deontological pooling at c_{max} by more (lower) types. What becomes ambiguous is the comparison at low μ (H2), which depends in very complex ways on the agent's type (low enough v's always contribute more under *DE,* high enough ones under *MPL*), the cost level c, and the entire distributions $G(c)$ and $F(v)$.

References

Andersen, Steffen, Glenn W. Harrison, Morten Igel Lau, and E. Elisabet Rutström (2006). "Elicitation Using Multiple Price List Formats". *Experimental Economics* 9 (4): 383–405. [16, 18]

Andreoni, James (1989). "Giving with Impure Altruism: Applications to Charity and Ricardian Equivalence". *Journal of Political Economy* 97 (6): 1447–58. [5]

Andreoni, James (1990). "Impure Altruism and Donations to Public Goods: A Theory of Warm-Glow". *Economic Journal* 100 (401): 464–77. [5]

Andreoni, James, and B. Douglas Bernheim (2009). "Social Image and the 50-50 Norm: A Theoretical and Experimental Analysis of Audience Effects". *Econometrica* 77 (5): 1607–36. [30]

Andreoni, James, and Ragan Petrie (2004). "Public Goods Experiments without Confidentiality: A Glimpse into Fund-Raising". *Journal of Public Economics* 88 (7–8): 1605–23. [33]

Ariely, Dan, Anat Bracha, and Stephan Meier (2009). "Doing Good or Doing Well? Image Motivation and Monetary Incentives in Behaving Prosocially". *American Economic Review* 99 (1): 544–55. [4, 31]

Ashraf, Nava, Oriana Bandiera, and B. Kelsey Jack (2014). "No Margin, No Mission? A Field Experiment on Incentives for Public Service Delivery". *Journal of Public Economics* 120: 1–17. [4]

Awad, Edmond, Sohan Dsouza, Richard Kim, Jonathan Schulz, Joseph Henrich, Azim Shariff, Jean François Bonnefon, and Iyad Rahwan (2018). "The Moral Machine Experiment". *Nature* 563 (7729): 59–64. [20]

Baillon, Aurelien, Yoram Halevy, and Chen Li (2022). "Randomize at Your Own Risk: On the Observability of Ambiguity Aversion". *Econometrica* 90 (3): 1085–107. [4]

Banks, Jeffrey S., Colin F. Camerer, and David Porter (1994). "An Experimental Analysis of Nash Refinements in Signaling Games". *Games and Economic Behavior* 6 (1): 1–31. [5]

Barmettler, Franziska, Ernst Fehr, and Christian Zehnder (2012). "Big Experimenter Is Watching You! Anonymity and Prosocial Behavior in the Laboratory". *Games and Economic Behavior* 75 (1): 17–34. [13]

Bartling, Björn, Vanessa Valero, Roberto A. Weber, and Lan Yao (1, 2024). "Public Discourse and Socially Responsible Market Behavior". *American Economic Review* 114 (10): 3041–74. [4]

Bašić, Zvonimir, and Simone Quercia (2022). "The Influence of Self and Social Image Concerns on Lying". *Games and Economic Behavior* 133: 162–69. [32]

Bénabou, Roland, Armin Falk, and Luca Henkel (2026). "Ends versus Means: Kantians, Utilitarians and Moral Decisions". *Working Paper*, [5, 20]

Bénabou, Roland, and Luca Henkel (2025). "Identity as self-image". *Working Paper*, [19]

Bénabou, Roland, and Jean Tirole (2006). "Incentives and Prosocial Behavior". *American Economic Review* 96 (5): 1652–78. [4]

Bénabou, Roland, and Jean Tirole (2011). "Identity, Morals, and Taboos: Beliefs as Assets". *Quarterly Journal of Economics* 126 (2): 805–55. [4]

Bénabou, Roland, and Jean Tirole (2025). "Laws and Norms". *Journal of Political Economy* forthcoming: [4]

Berry, James, Greg Fischer, and Raymond Guiteras (2020). "Eliciting and Utilizing Willingness to Pay: Evidence from Field Trials in Northern Ghana". *Journal of Political Economy* 128 (4): 1436–73. [5]

Birke, David J (2023). “Anti-Bunching: A New Test for Signaling Motives in Prosocial Behavior”. *Working Paper*, [35]

Birnbaum, Michael H (1992). “Violations of Monotonicity and Contextual Effects in Choice-Based Certainty Equivalents”. *Psychological Science* 3 (5): 310–15. [16]

Bock, Olaf, Ingmar Baetge, and Andreas Nicklisch (2014). “Hroot: Hamburg Registration and Organization Online Tool”. *European Economic Review* 71: 117–20. [13]

Bos, Olivier, and Martin Pollrich (2025). “Auctions with Signaling Bidders: Optimal Design and Information Disclosure”. *Games and Economic Behavior* 151: 95–107. [4]

Bos, Olivier, and Tom Truyts (2023). “Entry in First-Price Auctions with Signaling”. *International Journal of Game Theory* 52: 423–50. [4]

Bosch-Rosa, Ciril, Daniele Nosenzo, and Levent Neyse (2025). “Observability, Honesty, and the Social Image Costs of Lying”. *Working Paper*, [32]

Bradley, Alex, Claire Lawrence, and Eamonn Ferguson (28, 2018). “Does Observability Affect Prosociality?” *Proceedings of the Royal Society B: Biological Sciences* 285 (1875): 20180116. [2, 29]

Brandts, Jordi, and Gary Charness (2011). “The Strategy versus the Direct-Response Method: A First Survey of Experimental Comparisons”. *Experimental Economics* 14 (3): 375–98. [4]

Brandts, Jordi, and Charles A. Holt (1992). “An Experimental Test of Equilibrium Dominance in Signaling Games”. *American Economic Review* 82 (5): 1350–65. [5]

Brock, J. Michelle, Andreas Lange, and Erkut Y. Ozbay (2013). “Dictating the Risk: Experimental Evidence on Giving in Risky Environments”. *American Economic Review* 103 (1): 415–37. [5]

Burchardi, Konrad B., Jonathan De Quidt, Selim Gulesci, Benedetta Lerva, and Stefano Tripodi (2021). “Testing Willingness to Pay Elicitation Mechanisms in the Field: Evidence from Uganda”. *Journal of Development Economics* 152 (9): 102701. [18]

Bursztyn, Leonardo, Bruno Ferman, Stefano Fiorin, Martin Kanz, and Gautam Rao (2018). “Status Goods: Experimental Evidence from Platinum Credit Cards”. *Quarterly Journal of Economics* 133 (3): 1561–95. [4]

Bursztyn, Leonardo, and Robert Jensen (2017). “Social Image and Economic Behavior in the Field: Identifying, Understanding, and Shaping Social Pressure”. *Annual Review of Economics* 9: 131–53. [1]

Carattini, Stefano, Kenneth Gillingham, Xiangyu Meng, and Erez Yoeli (2024). “Peer-to-Peer Solar and Social Rewards: Evidence from a Field Experiment”. *Journal of Economic Behavior & Organization* 219: 340–70. [36]

Chakraborty, Anujit, and Luca Henkel (2024). “The Role of Interpersonal Uncertainty in Prosocial Behavior”. *Working Paper*, [5]

Charness, Gary, Uri Gneezy, and Brianna Halladay (2016). “Experimental Methods: Pay One or Pay All”. *Journal of Economic Behavior and Organization* 131: 141–50. [4]

Charness, Gary, Uri Gneezy, and Alex Imas (2013). “Experimental Methods: Eliciting Risk Preferences”. *Journal of Economic Behavior and Organization* 87: 43–51. [4]

Chen, Daniel L., and Martin Schonger (2016). “A Theory of Experiments: Invariance of Equilibrium to the Strategy Method of Elicitation and Implications for Social Preferences”. *TSE Working Paper, no. 16-724*, [4]

Chen, Daniel L., and Martin Schonger (2022). "Social Preferences or Sacred Values? Theory and Evidence of Deontological Motivations". *Science Advances* 8 (19): eabb3925. [5]

Chen, Daniel L., Martin Schonger, and Chris Wickens (2016). "oTree-An Open-Source Platform for Laboratory, Online, and Field Experiments". *Journal of Behavioral and Experimental Finance* 9: 88–97. [13]

Cohen, Jonathan, Keith Marzilli Ericson, David Laibson, and John Myles White (2020). "Measuring Time Preferences". *Journal of Economic Literature* 58 (2): 299–347. [4]

Cole, Shawn, A Nilesh Fernando, Daniel Stein, and Jeremy Tobacman (2020). "Field Comparisons of Incentive-Compatible Preference Elicitation Techniques". *Journal of Economic Behavior and Organization* 172: 33–56. [5]

Cox, James C., Vjollca Sadiraj, and Ulrich Schmidt (2015). "Paradoxes and Mechanisms for Choice under Risk". *Experimental Economics* 18 (2): 215–50. [4]

Cubitt, Robin P., Chris Starmer, and Robert Sugden (1998). "On the Validity of the Random Lottery Incentive System". *Experimental Economics* 1: 115–31. [16]

Dana, Jason, Roberto A. Weber, and Jason Xi Kuang (2007). "Exploiting Moral Wiggle Room: Experiments Demonstrating an Illusory Preference for Fairness". *Economic Theory* 33 (1): 67–80. [4]

Dannenberg, Astrid, Charlotte Klatt, and Eva Weingärtner (2024). "The Effects of Social Norms and Observability on Food Choice". *Food Policy* 125: 102621. [31]

DellaVigna, Stefano, John A. List, and Ulrike Malmendier (2012). "Testing for Altruism and Social Pressure in Charitable Giving". *Quarterly Journal of Economics* 127 (1): 1–56. [4]

Exley, Christine (2018). "Incentives for Prosocial Behavior: The Role of Reputations". *Management Science* 64 (5): 2460–71. [35]

Exley, Christine L. (2016). "Excusing Selfishness in Charitable Giving: The Role of Risk". *Review of Economic Studies* 83 (2): 587–628. [4]

Falk, Armin (2021). "Facing Yourself - A Note on Self-Image". *Journal of Economic Behavior and Organization* 186: 724–34. [4]

Falk, Armin, and Thomas Graeber (2020). "Delayed Negative Effects of Prosocial Spending on Happiness". *Proceedings of the National Academy of Sciences* 117 (12): 6463–68. [11, 12]

Falk, Armin, Thomas Neuber, and Nora Szech (2020). "Diffusion of Being Pivotal and Immoral Outcomes". *Review of Economic Studies* 87 (5): 2205–29. [4, 5]

Feddersen, Timothy, Sean Gailmard, and Alvaro Sandroni (2009). "Moral Bias in Large Elections: Theory and Experimental Evidence". *American Political Science Review* 103 (2): 175–92. [4]

Filiz-Ozbay, Emel, and Erkut Y. Ozbay (2014). "Effect of an Audience in Public Goods Provision". *Experimental Economics* 17 (2): 200–14. [33]

Foot, Philippa (1967). "The Problem of Abortion and the Doctrine of the Double Effect". *Oxford Review* 5: 5–15. [20]

Friedrichsen, Jana, and Dirk Engelmann (2018). "Who Cares about Social Image?" *European Economic Review* 110: 61–77. [30]

Galeotti, Fabio, Valeria Maggian, and Marie Claire Villeval (2021). "Fraud Deterrence Institutions Reduce Intrinsic Honesty". *Economic Journal* 131 (638): 2508–28. [4]

Garcia, Thomas, Sébastien Massoni, and Marie Claire Villeval (2020). “Ambiguity and excuse-driven behavior in charitable giving”. *European Economic Review* 124: 103412. [4]

Giovannoni, Francesco, and Miltiadis Makris (2014). “Reputational Bidding”. *International Economic Review* 55 (3): 693–710. [4]

Gneezy, Uri, Elizabeth A. Keenan, and Ayelet Gneezy (2014). “Avoiding Overhead Aversion in Charity”. *Science* 346 (6209): 632–35. [5]

Goeree, Jacob K. (2003). “Bidding for the Future: Signaling in Auctions with an Aftermarket”. *Journal of Economic Theory* 108 (2): 345–64. [4]

Goeree, Jacob K., Charles A. Holt, and Susan K. Laury (2002). “Private Costs and Public Benefits: Unraveling the Effects of Altruism and Noisy Behavior”. *Journal of Public Economics* 83 (2): 255–76. [5]

Grossman, Zachary (2015). “Self-Signaling and Social-Signaling in Giving”. *Journal of Economic Behavior and Organization* 117: 26–39. [4]

Grossman, Zachary, and Joël J. van der Weele (2017). “Self-Image and Willful Ignorance in Social Decisions”. *Journal of the European Economic Association* 15 (1): 173–217. [4]

Hey, John D, and Jinkwon Lee (2005). “Do Subjects Separate (or Are They Sophisticated)?” *Experimental Economics* 8 (3): 233–66. [16]

Hofmeier, Jana, and Louis Strang (6, 2023). “Image Concerns and the Dynamics of Prosocial Behavior”. *Frontiers in Behavioral Economics* 2: 1220007. [20, 34]

Jee, Edward, Anne Karing, and Karim Naguib (2025). “Social Image and the Social Multiplier: Experimental Evidence from Community Deworming in Kenya”. *Working Paper*, [36]

Kagel, John H., Ronald M. Harstad, and Dan Levin (1987). “Information Impact and Allocation Rules in Auctions with Affiliated Private Values: A Laboratory Study”. *Econometrica* 55 (6): 1275–304. [18]

Karing, Anne (9, 2024). “Social Signaling and Childhood Immunization: A Field Experiment in Sierra Leone”. *The Quarterly Journal of Economics* 139 (4): 2083–133. [36]

Karlan, Dean, and Margaret A. McConnell (2014). “Hey Look at Me: The Effect of Giving Circles on Giving”. *Journal of Economic Behavior & Organization* 106: 402–12. [30]

Kolappan, C., R. Subramani, V. Kumaraswami, T. Santha, and P. R. Narayanan (2008). “Excess Mortality and Risk Factors for Mortality among a Cohort of TB Patients from Rural South India”. *International Journal of Tuberculosis and Lung Disease* 12 (1): 81–86. [11]

Ledyard, John O. (1995). “Public Goods: A Survey of Experimental Research”. In *The Handbook of Experimental Economics*. Alvin E. Rot and John H. Kagel, ed. vol. 1, Princeton University Press, 111–94. [5]

Li, Zhihua, Julia Müller, Peter P. Wakker, and Tong V. Wang (2017). “The Rich Domain of Ambiguity Explored”. *Management Science* 64 (7): 3227–40. [18]

Miller, Klaus M., Reto Hofstetter, Harley Krohmer, and Z. John Zhang (2011). “How Should Consumers’ Willingness to Pay Be Measured? An Empirical Comparison of State-of-the-Art Approaches”. *Journal of Marketing Research* 48 (1): 172–84. [5]

Pallottini, Ashton (2025). “Onset of Offsets: The Evolution of Social Rewards in Demand for Environmental Goods”. *Working Paper*, [36]

Rege, Mari, and Kjetil Telle (2004). “The Impact of Social Approval and Framing on Cooperation in Public Good Situations”. *Journal of Public Economics* 88 (7–8): 1625–44. [35]

Savikhin Samek, Anya, and Roman M. Sheremeta (2014). “Recognizing Contributors: An Experiment on Public Goods”. *Experimental Economics* 17 (4): 673–90. [34]

Selten, Reinhard (1967). “Die Strategiemethode Zur Erforschung Des Eingeschränkt Rationalen Verhaltens Im Rahmen Eines Oligopolexperiments”. In *Beiträge Zur Experimentellen Wirtschaftsforschung*. H. Sauermann, ed. Tübingen: Mohr, 136–68. [4]

Simonson, Itamar (1989). “Choice Based on Reasons: The Case of Attraction and Compromise Effects”. *Journal of Consumer Research* 16 (2): 158. [16]

Soetevent, Adriaan R. (2005). “Anonymity in Giving in a Natural Context—a Field Experiment in 30 Churches”. *Journal of Public Economics* 89 (11–12): 2301–23. [34]

Starmer, Chris, and Robert Sugden (1991). “Does the Random-Lottery Incentive System Elicit True Preferences? An Experimental Investigation”. *American Economic Review* 81 (4): 971–78. [16]

Straetemans, Masja, Philippe Glaziou, Ana L. Bierrenbach, Charalambos Sismanidis, and Marieke J. van der Werf (2011). “Assessing Tuberculosis Case Fatality Ratio: A Meta-Analysis”. *PLoS ONE* 6 (6): [11]

Teyssier, Sabrina, Fabrice Etilé, and Pierre Combris (2015). “Social- and Self-Image Concerns in Fair-Trade Consumption: Evidence from Experimental Auctions for Chocolate”. *Working Paper*, [31]

Thompson, Judith Jarvis (1976). “Killing, Letting Die, and the Trolley Problem”. *Monist* 59 (2): 204–17. [20]

Tiemersma, Edine W., Marieke J. van der Werf, Martien W. Borgdorff, Brian G. Williams, and Nico J.D. Nagelkerke (2011). “Natural History of Tuberculosis: Duration and Fatality of Untreated Pulmonary Tuberculosis in HIV Negative Patients: A Systematic Review”. *PLoS ONE* 6 (4): [11]

Van Leeuwen, Boris, and Ingela Alger (1, 2024). “Estimating Social Preferences and Kantian Morality in Strategic Interactions”. *Journal of Political Economy Microeconomics* 2 (4): 665–706. [5]

SUPPLEMENTARY APPENDIX

A Details on the meta-analysis

This section provides more details on the meta-analysis presented in Figure 1. For panel A, we searched for economic papers that (i) feature an explicit treatment manipulation that varies the observability of choices to others, (ii) study its impact on prosocial behavior. We define prosocial behavior broadly as a choice that arguably has consequences for individuals other than the decision-maker. We explicitly included working papers, not only published work. In total, we identified 20 papers. When a paper included multiple observability treatments, we tried to include as many as possible. However, in several instances it was not possible to recover the effect size from the paper.[19] In total, we were able to recover data from 42 treatment comparisons. Table A.1 lists each included paper, together with information about the sample size and the treatments that were included in the figure, as well as descriptions of the treatment and decision. For each treatment comparison, we calculated the standardized effect size using Fisher's z for the effect of increased observability on the treatment comparison's main outcome.

For panel B, we use the data provided by Bradley et al. (2018). They conducted a meta-analysis of studies that vary observability and have prosociality as an outcome measure. Their sample includes mostly - but not exclusively - papers in psychology. For observability variations, they include treatments that either reveal information about an individual's identity or their behavior. Regarding the outcome, they include treatments whose outcome measures are intentions or behaviors that benefit an individual, group, organization, or society. This yields 112 papers and 234 treatment comparisons. For more details, see Bradley et al. (2018).

[19] For instance, some papers include only marginal effects and do not provide access to their experimental data.

Table A.1: Papers included in the meta-analysis

Paper	n	Treatments	Treatment description	Decision description
Friedrichsen and Engelmann (2018)	121	Public vs. Private	Subjects announce their decision to everyone in the room versus remaining private.	Subjects state their WTP for both a fair trade and a conventional chocolate bar. Prices range from 0 to 2€ in 0.01€ increments. The outcome is subjects premium for the fair trade chocolate.
Andreoni and Bernheim (2009)	236	p≥0.25 versus $p = 0$ for $x_0 = 0$	Subjects' choices are either fully attributable to them or can be partially attributed to chance (nature).	Dictator game with an endowment of $20 and another subject as recipient.
Karlan and McConnell (2014)	4,168 (field); 94 (lab)	Treatment 100 circle vs. Control; Treatment 500 circle vs. Control; Treatment both 100 and 500 circles vs. Control; Signaling vs. Image	Field experiment: Alumni solicited by phone with scripts that vary whether donations above $100 and/or $500 are publicly recognized in a newsletter or kept private. Lab experiment: Subjects' choices with their name written on room's blackboard or kept private.	Field experiment: How much to donate to Yale service club. We use their log(1+gift amount) outcome. Lab experiment: Dictator game with endowment of $5 and a charity as recipient.

Table A.1: Papers included in the meta-analysis (continued)

Paper	n	Treatments	Treatment description	Decision description
Ariely et al. (2009)	161; 161; 151	Public vs. Private (controlling for individual perception); Public vs. Private (controlling for the consensus); Public vs. Private	Lab experiment: Subjects disclose to other subjects their assigned charity, whether they receive private monetary incentives, and the amount donated to the charity, versus remaining anonymous. Field experiment: Subjects bike either in public view or in a private location of a gym.	Lab experiment: Subjects choose how much effort to exert in a five-minute real-effort task, where repeated key presses generate charitable donations under a decreasing payment schedule. Field experiment: Subjects choose how much effort to exert by cycling on stationary bikes for up to ten minutes at a donation rate of $1 per mile.
Dannenberg et al. (2024)	931	Public vs. Private	Subjects' names and choices or only their names are announced aloud.	Subjects choose between a voucher for a vegan, vegetarian, or meat sandwich. We use the meatless versus meat outcome.
Teyssier et al. (2015)	110	Public vs. Private	Subjects' choices are visible to others in the same session or kept private.	Subjects state their WTP for both a fair trade and a conventional chocolate bar. Prices range from 1.20 to 5.20€ in 0.10€ increments.

Table A.1: Papers included in the meta-analysis (continued)

Paper	n	Treatments	Treatment description	Decision description
Bašić and Quercia (2022)	531; 306	Social vs. Control (Experiment 1); Social vs. Self-Call (Experiment 2)	Experiment 1: Subjects are observed by another subject while reporting their decision versus not being observed. Experiment 2: Subjects are observed through an online video call by the experimenter while reporting their decision versus not observed.	Die rolling lying game: subjects privately roll a die and decide whether to report the true number or not. Higher reported numbers yield higher payoffs.
Bosch-Rosa et al. (2025)	1,183	Observable vs. Unobservable	Observability is manipulated through two reporting tasks. In one task, the experimenter cannot observe the true realization of the random variable reported by subjects. In the other task, both the realization and the report are observable.	Adaptation of the die rolling lying game: subjects privately observe the realization of a 10-state random variable and decide whether to report the true number or not. Higher reported realizations yield higher payoffs.

Table A.1: Papers included in the meta-analysis (continued)

Paper	n	Treatments	Treatment description	Decision description
Andreoni and Petrie (2004)	80; 80; 80	Information-and-Photos vs. Information; Information-and-Photos vs. Photos; Optional-Reporting vs. Information-and-Photos	Subjects' contributions may be private, publicly revealed without identification, or linked to digital photographs. In the Optional-Reporting treatment, subjects choose between a broadcast public good with publicly identified contributions and an anonymous public good where only total contributions are observed.	Repeated public goods game with an endowment of 20 tokens, a group size of five and 40 rounds. Tokens invested in the private good pay \$0.02 to the contributor, while tokens invested in a public good pay \$0.01 to each group member.
Filiz-Ozbay and Ozbay (2014)	84; 60; 80; 64	EOS vs. ES; EOR vs. ER; OS vs. S; OR versus R	Subjects are observed by another person while making their decision versus not observed.	Public goods game with an endowment of \$5 and a group size of four. In the effort treatments, subjects decide whether to pay \$0.50 to attempt effort tasks (adding five two-digit numbers), with each correct task generating \$1 for each group member. In non-effort treatments, subjects allocate the endowment between a private account and a group account, which is doubled and shared equally.

Table A.1: Papers included in the meta-analysis (continued)

Paper	n	Treatments	Treatment description	Decision description
Savikhin Samek and Sheremeta (2014)	3,200	ALL vs. NONE; TOP vs. NONE; BOTTOM vs. NONE	Contributors are publicly identified by name and photo next to their contribution depending on treatment: All contributors are identified, only the highest contributors, only the lowest contributors, or no information is revealed.	Repeated public goods game with an endowment of 80 tokens (=$4), a group size of five and 20 rounds. Contributions are multiplied by 0.4 and evenly distributed among group members.
Soetevent (2005)	1,582	Basket vs. Bag (1st offering); Basket vs. Bag (2nd offering)	Donations are collected using closed bags versus open baskets.	During mass, donations are collected by passing the bag or basket along the rows, and church attendees decide how much to donate.
Hofmeier and Strang (2023)	203; 203; 77; 77	Public vs. Private (Exp A, Stage 1); Stage 2 after public exposure (Exp A); Public vs. Private (Exp B, Stage 1); Stage 2 after public exposure (Exp B)	Experiment A: Subjects' donation choice is either observed by an observer or made privately. Experiment B: Subjects' donation choice is either publicly announced to others of the same session or kept private.	Experiment A: Subjects count zeros to generate a charitable donation, with earnings determined by a decreasing piece rate. In Stage 2, subjects decide how much of a 5 € endowment to donate in a double-blind dictator game. Experiment B: In both Stage 1 and Stage 2, subjects count zeros to generate the donation.

Table A.1: Papers included in the meta-analysis (continued)

Paper	n	Treatments	Treatment description	Decision description
Birke (2023)	2,131	No badge vs. Public badge	Subjects' choices are displayed through a personal badge that is shown to at least one other subjects versus remain private.	Subjects perform a real-effort transcription task consisting of copying 38 Greek letters from an image. Each completed task generates 0.08 € for a charity.
Exley (2018)	800; 130	Public vs. Private reputation (online); Public vs. Private reputation (lab)	Online: Subjects choice is either observed by a matched panel member or kept private. Lab: Information about subjects' past volunteer behavior is either observable to a matched panel member or kept private.	Online: Subjects count zeros in seven tables. First, they decide whether to complete an unpaid task that generates a \$10 donation to the American Red Cross. Lab: Subjects click a button on an electronic tally counter for eight minutes. Every five clicks generate a \$0.01 donation to the American Red Cross and a \$0.01 payment to the subject.
Rege and Telle (2004)	40	Approval vs. No-Approval (Non Associative framing); Approval vs. No-Approval (Associative framing)	Subjects publicly reveal their choices or not.	Public goods game with an endowment of 150 kroner (≈\$17) and a group size of ten. Money to the group is doubled and then divided equally among all group members.

Table A.1: Papers included in the meta-analysis (continued)

Paper	n	Treatments	Treatment description	Decision description
Karing (2024)	4,897	Vaccine-5 signal vs. Control (timely v5); Vaccine-4 signal vs. Control (timely v5); Vaccine-4 signal vs. Control (timely v4); Vaccine-5 signal vs. Control (timely v4)	Parents receive color-coded bracelets that either signal whether their child has completed vaccinations on schedule or contain no signal.	Parents decide whether to complete vaccinations on schedule for their children or not.
Pallottini (2025)	2,305	Public vs. Private	Subjects' choices are posted on a website together with their names or kept private.	Subjects decide whether to allocate part of a bonus payment to purchase a carbon offset that mitigates 145 lbs of emissions or to keep the full bonus.
Carattini et al. (2024)	176,546	Green Reports vs. No Reports	Subjects' choices are publicly shareable online through green reports or remain private.	Subjects decide whether to click on an online advertisement promoting participation in a peer-to-peer solar energy program.
Jee et al. (2025)	9,805	Bracelet vs. Control	Subjects receive a bracelet signaling successfully completed deworming or no signal is provided.	Individuals decide whether to attend deworming offered in their community.

B Additional Tables

Table B.1: Regression analyses of the effect of the elicitation method on prosocial behavior excluding subjects failing comprehension questions

Panel A:				
Dependent variable:	Choice to Save a Life (vs. $100€$)			
	Low Image		*High Image*	
	(1)	(2)	(3)	(4)
MPL	-0.115^{**}	-0.113^{**}	0.101^{**}	0.098^{*}
	(0.054)	(0.054)	(0.051)	(0.051)
Constant (*DE*)	0.595^{***}	0.638^{***}	0.616^{***}	0.601^{***}
	(0.039)	(0.050)	(0.037)	(0.047)
Controls		X		X
Observations	338	338	338	338

Panel B:		
Dependent variable:	Choice to Save a Life (vs. $100€$)	
	(1)	(2)
MPL	-0.115^{**}	-0.103^{*}
	(0.054)	(0.053)
High Image	0.021	0.038
	(0.054)	(0.052)
MPL X High Image	0.216^{***}	0.202^{***}
	(0.074)	(0.073)
Constant (*DE Low Image*)	0.595^{***}	0.597^{***}
	(0.039)	(0.044)
Controls		X
Observations	676	676

Notes: The table replicates Table 1 using the sample of subjects that make at most one mistake in the comprehension question quiz.

C Extension: continuum of types

Let agents now differ in their prosociality v, according to a uniform distribution on $[0, v_{max}]$, where $v_{max} < \infty$. Denote the average v by $\bar{v} = v_{max}/2$. Besides their intrinsic motivation v to do good and their extrinsic motivation (the cost of the action), they also care about their reputational payoff $R \equiv \mu\widehat{v}$, where $\widehat{v}$ is their reputation and μ the (known) intensity of their image concerns. For notational simplicity, we normalize $e = 1$.

C.1 Direct elicitation

The supply of prosociality under *DE* is $\bar{a}^{DE}(c) \equiv 1 - \frac{v^{DE}(c)}{v_{max}}$, where the unique cutoff $v^{DE}(c)$, when interior, solves

$$v^{DE}(c) - c + \mu\Delta = 0$$

and $\Delta \equiv \mathbb{E}(v|v > v^*) - \mathbb{E}(v|v < v^*) = \mu\bar{v}$, for all $v^* \in [0, v_{max}]$.[20] We can define v's implicit bid as

$$\begin{aligned} b^{DE}(v) &\equiv \max\left\{c \mid a^{DE}(v,c) = 1\right\} \\ &= \max\left\{c \mid v \geq v^{DE}(c)\right\} = \min\{v + \mu\bar{v}, c_{max}\} \end{aligned}$$

and the image-induced inflation in the willingness to pay (henceforth, WTP) as

$$x^{DE}(v) \equiv b^{DE}(v) - v = \min\{\mu\bar{v}, c_{max} - v\}.$$

C.2 Multiple-Price-List

We consider a uniform cost distribution (as is the case in most experiments, including the one conducted in this paper) with $g(c) = 1$, meaning that we normalize $c_{max} = 1$. An agent's utility when having type v and bidding b is:

$$U(v,b) \equiv \int_0^b (v - \tilde{c})d\tilde{c} + \mu R(b) = vb - \frac{b^2}{2} + \mu R(b)$$

We look for a *separating equilibrium* on an interval $[0, v^\dagger]$ combined with pooling at $c_{\max} = 1$ on $(v^\dagger, v_{max}]$ (a fully separating equilibrium corresponds to $v^\dagger = v_{max}$).

Let $b(v)$ stands for the equilibrium bid of type v. In the separating part of the equilibrium

$$b(0) = 0,$$

and $b(v) = \max_b\{vb - \frac{b^2}{2} + \mu\hat{v}(b)\}$, which satisfies:

$$\frac{db}{dv}[b(v) - v] = \mu.$$

Finally, types $v > v^\dagger$ select $b(v) = 1$: they behave in an observationally deontological (or price insensitive) way, choosing $a = 1$ at any $c \leq c_{max}$ At the jump point $v^\dagger$ (when interior), this type gains extra reputation $\mu[M^\dagger(v^\dagger) - v^\dagger] = \mu\left[\frac{1-v^\dagger}{2}\right]$ by pooling with higher types in this way, but increases the cost of signaling from $\left[b(v^\dagger) - v^\dagger\right]^2/2$ to $\left[1 - v^\dagger\right]^2/2$.

And so,

$$(1 - v^\dagger)^2 - (b(v^\dagger) - v^\dagger)^2 = \mu(v_{max} - v^\dagger). \tag{10}$$

[20]Corner solutions are $v^{DE}(c) = 0$ when $c \leq \mu\bar{v}$, and $v^{DE}(c) = v_{max}$ when $c \geq v_{max} + \mu\bar{v}$.

This jump point could be 0 (everyone mimics “Kantian” behavior"), or could be v_{max} when the equilibrium is fully separating. We will consider all of these cases when comparing *DE* and *MPL*. Let

$$x^{MPL}(v) \equiv b(v) - v$$

denote type v's inflation of WTP. Solving for the differential equation yields, for $v < v^{\dagger}$,

$$x^{MPL}(v) = \mu\big[1 + W(-e^{-1-v/\mu})\big],$$

where $W : [-e^{-1}, \infty) \to [-1, \infty)$ is the Lambert W-function, i.e., the inverse function of xe^x. Since $W(-e^{-1}) = -1$, we have $x^{MPL}(0) = 0$. Moreover, $x^{MPL}(v)$ goes to μ as v goes large, because $W(0) = 0$. Note that $x^{MPL}(v)$ is strictly increasing in v, and so a fortiori is $b^{MPL}(v)$. Moreover, the inflation is increasing in μ:

$$\frac{\partial x^{MPL}}{\partial \mu} = 1 + w + \frac{v}{\mu} \cdot \frac{w}{1+w} = \frac{1 + w - w\ln(-w)}{1+w}$$

where $w = W(-e^{-1-v/\mu}) \in [-1, 0)$, and using

$$we^w = -e^{-1-v/\mu} \implies \frac{v}{\mu} = -\big(\ln(-w) + w + 1\big).$$

The derivative of x^{MPL} with respect to μ is positive because the numerator $1+w-w\ln(-w)$ equals 0 at $w = -1$ and is increasing in w (the denominator is obviously positive).

C.3 Comparing *DE* and *MPL*.

Depending on the magnitudes of v_{max} and μ, different cases may happen.

Case 1: $\frac{2}{\mu} \leq v_{max} \Leftrightarrow c_{max} \leq \mu\bar{v}$. Behavior of all types is observationally deontological, under both *DE* and *MPL:*

$$\forall v : \quad b^{MPL}(v) = b^{DE}(v) = 1.$$

Case 2: $\frac{1}{\mu} \leq v_{max} < \frac{2}{\mu} \Leftrightarrow \bar{c} \leq \mu\bar{v} < c_{max}$. Behavior is always observationally deontological under *MPL,* but not under *DE:*

$$\forall v : \quad b^{MPL}(v) = 1 \quad \text{and} \quad b^{DE}(v) = \min\{v + \mu\bar{v}, 1\}.$$

Case 3: $\leq v_{max} < \frac{1}{\mu}$ (this can only happen when $\mu < 0.5$). The two curves do not intersect because $\Delta \geq 1$. Moreover, b^{MPL} jumps to c_{max} at $v^{\dagger} \in (0, v_{max})$, but there could be two cases:

(a) If $v^{\dagger} < 1 - \mu\bar{v}$ (that is, $b^{DE}(v^{\dagger}) = v^{\dagger} + \mu\bar{v} < 1$), then b^{MPL} jumps above b^{DE} at $v^{\dagger}$.

(b) Otherwise, b^{MPL} jumps to c_{max} when b^{DE} is already there. Thus b^{MPL} is below b^{DE} everywhere.

Proof. Suppose first that the two curves intersect. Then we have:

$$\mu\bar{v} = \mu + \mu W(-e^{-1-v/\mu}) \Rightarrow W(-e^{-1-v/\mu}) = \bar{v} - 1 = \frac{v_{max}}{2} - 1 \geq 0,$$

while $W(-e^{-1-v/\mu})$ is always negative, a contradiction. Therefore, the two curves cannot intersect when $\Delta \geq 1$.

Moreover, MPL is fully separating if and only if

$$b(v_{max}) = v_{max} + \mu + \mu W(e^{-1-v_{max}/\mu}) \leq 1.$$

But here, $v_{max} \geq 2$ and we know that $\mu + \mu W(e^{-1-v/\mu}) > 0$ for all v. Thus, MPL cannot be fully separating, and $v^\dagger < v_{max}$. Furthermore, we know that

$$(1 - v^\dagger)^2 - (b(v^\dagger) - v^\dagger)^2 = 2\mu(\bar{v} - \frac{v^\dagger}{2}),$$

which shows that $v^\dagger \neq 0$: $v^\dagger = 0$ would imply that $1 = 2\mu\bar{v} = \mu v_{max}$, a contradiction.

Therefore, because DE reaches $c_{max} = 1$ at $1 - \mu\bar{v}$, either $v^\dagger < 1 - \mu\bar{v}$, and MPL jumps above b^{DE} at $v^\dagger$, or MPL jumps to c_{max} when DE is already there. Q.E.D.

Case 4: $v_{max} < \min\{2, \frac{1}{\mu}\}$
Denote the potential crossing point by $\hat{v} = -\mu\big[\ln(1 - \bar{v}) + \bar{v}\big] > 0$. How does $\hat{v}$ compare with $v^\dagger$, where MPL jumps?

(a) If $\hat{v} \leq v^\dagger$, then two curves intersect at $\hat{v}$, and we have

$$b^{MPL}(\hat{v}) = b^{DE}(\hat{v}) = -\mu\ln(1 - \bar{v}).$$

This case includes the situation where MPL is fully separating. Indeed, whenever MPL is fully separating the two curves intersect and none of them reaches c_{max}.

(b) If $\hat{v} > v^\dagger$, b^{MPL} jumps above b^{DE} at $v^\dagger$, and they do not intersect.

Proof. The only claim that we have to prove in Case 4 is that: *whenever MPL is fully separating the two curves intersect and none of them reaches c_{max}*. Note that MPL can be fully separating only under Case 4's conditions, i.e. $v_{max} < \min\{2, \frac{1}{\mu}\}$. Moreover, when MPL is fully separating, $v_{max} \leq b(v_{max}) \leq 1$, and so, $\bar{v} \leq 0.5$. Considering these, we next prove that $\hat{v} = -\mu\big(\ln(1-\bar{v}) - \bar{v}\big) < 2\bar{v} = v_{max}$ for any $\bar{v} \leq 0.5$. Since $\mu < \frac{1}{v_{max}} = \frac{1}{2\bar{v}}$ and $-\big(\ln(1-\bar{v}) + \bar{v}\big) > 0$, we have:

$$\hat{v} = -\mu\big(\ln(1-\bar{v}) + \bar{v}\big) < \frac{-\ln(1-\bar{v})}{2\bar{v}} - \frac{1}{2}.$$

To prove that

$$\frac{-\ln(1-\bar{v})}{2\bar{v}} - \frac{1}{2} < 2\bar{v},$$

we show:

$$h(\bar{v}) \equiv -\ln(1-\bar{v}) - \bar{v} - 4\bar{v}^2 \leq 0 \quad \forall \bar{v} \in [0, 0.5].$$

This is true because $h(0) = h'(0) = 0$ and $h''(v) = \frac{1}{(1-\bar{v})^2} - 8 < 0 \; \forall \bar{v} \in [0, 0.5]$.

Finally, as we showed, the two curves intersect once and only once in the interval $[0, v_{max}]$, because MPL is fully separating. Moreover, after the intersection, $b^{DE}(v_{max})$ is below $b^{MPL}(v_{max})$. Therefore, $b^{DE}(v_{max}) < b^{MPL}(v_{max}) \leq 1$. Q.E.D.

Comparing DE and MPL for large and small μ. It is easy to see that, fixing v_{max}, for μ large (but not excessively large, i.e. we are in Case 2), all contribute under MPL, but not under DE. In contrast, as μ becomes small:

- If $v_{max} < 2$, Case 4(a) applies and b^{MPL} cuts b^{DE} at some v converging to 0 as μ goes to 0, and remains higher afterward.
- If $v_{max} \geq 2$, Case 3(a) applies and b^{MPL} is below b^{DE} before $v^\dagger$ and jumps above b^{DE} at $v^\dagger < 1 - \mu\bar{v}$.

Proof. Note that $v_{max} < \frac{1}{\mu}$ for small μ. When $v_{max} < 2$, Case 4 applies. Moreover, since $\hat{v}$ goes to 0 and $v^\dagger$ goes to $\min\{1, v_{max}\}$ as μ goes to 0, we have $\hat{v} < v^\dagger$ for small μ. On the other hand, when $v_{max} \geq 2$, Case 3 applies. Moreover, $v^\dagger < 1 - \mu\bar{v}$ for small enough μ, by the following lemma (putting $k = \bar{v}$ and $\alpha = 1$). We actually prove a stronger version than is needed here, as this will prove useful later on. Q.E.D.

Note that $v^\dagger$ depends on μ. Thus, we sometimes use $v^\dagger_\mu$ to emphasize this point.

Lemma 1. *When $2 \leq v_{max} < \frac{1}{\mu}$, for any k and α such that $\alpha > \frac{1}{2}$ or ($\alpha = \frac{1}{2}$ and $k < 2\bar{v} - 1$), there exists $\delta > 0$ such that for all $\mu < \delta$:*

$$v^\dagger_\mu < 1 - (k\mu)^\alpha.$$

Proof. Let us define

$$h(v, \mu) \equiv (1-v)^2 - (b(v) - v)^2 - 2\mu(\bar{v} - \frac{v}{2}).$$

Note that $v^\dagger_\mu$ must solves $h(v^\dagger_\mu, \mu) = 0$. Moreover, $h(0,\mu) = 1 - 2\mu\bar{v} > 0$, and $h(1,\mu) = -2\mu(\bar{v} - \frac{1}{2}) < 0$. Additionally,

$$\begin{aligned}\frac{\partial h(v,\mu)}{\partial v} &= -2(1-v) - 2(b'(v)-1)(b(v)-v) + \mu \\ &= -2(1-v) - 2\mu + 2(b(v)-v) + \mu = 2(b(v)-1) - \mu < 0.\end{aligned}$$

So $v^\dagger_\mu$ exists and we can prove that $v^\dagger_\mu < 1 - (k\mu)^\alpha$ by showing $h(1-(k\mu)^\alpha, \mu) < 0$. We have:

$$h(1-(k\mu)^\alpha, \mu) = (k\mu)^{2\alpha} - \Big(\mu + \mu W(-e^{-1-\frac{1}{\mu}+k^\alpha\mu^{\alpha-1}})\Big)^2 - 2\mu\bar{v} + \mu(1-(k\mu)^\alpha).$$

Therefore, denoting $w = W(-e^{-1-\frac{1}{\mu}+k^\alpha\mu^{\alpha-1}})$,

$$\begin{aligned}\frac{d}{d\mu}h(1-k\mu^\alpha,\mu) &= 2\alpha k^{2\alpha}\mu^{2\alpha-1} - 2(\mu+\mu w)\Big(1 + w + \mu\cdot\frac{1}{\mu^2}\big(1+(\alpha-1)(k\mu)^\alpha\big)\frac{w}{1+w}\Big) \\ &\qquad - 2\bar{v} + 1 - (1+\alpha)(k\mu)^\alpha \\ &= 2\alpha k^{2\alpha}\mu^{2\alpha-1} - 2(1+w)\Big(\mu + \mu w + \big(1+(\alpha-1)(k\mu)^\alpha\big)\frac{w}{1+w}\Big) \\ &\qquad - 2\bar{v} + 1 - (1+\alpha)(k\mu)^\alpha.\end{aligned}$$

Thus,

$$\frac{d}{d\mu}h(1-(k\mu)^\alpha,\mu)|_{\mu=0} = 1 - 2\bar{v} + 2\alpha k^{2\alpha}\mu^{2\alpha-1}|_{\mu=0} = \begin{cases} 1-2\bar{v} & \text{if } \alpha > \frac{1}{2} \\ 1 - 2\bar{v} + k & \text{if } \alpha = \frac{1}{2}\end{cases}.$$

This is negative since $\bar{v} > 1$, and when $\alpha = \frac{1}{2}$ we have $k < 2\bar{v} - 1$. Moreover, $h(1,0) = 0$. Therefore, for μ close enough to zero, $h(1-(k\mu)^\alpha, \mu) < 0$. Q.E.D.

Comparing average inflations. Let us also compute the average bid inflations from 0 to v, $A^{MPL}(v)$ and $A^{DE}(v)$. We do not multiply by the uniform density $f(v) = \frac{1}{v_{max}}$, which means we calculate averages times v_{max}. For $v \leq 1 - \mu\bar{v}$ we have:

$$A^{DE}(v) = \int_0^v x^{DE}(s)ds = \int_0^v \mu\bar{v}ds = \mu\bar{v}v.$$

Moreover, for $v \leq v^\dagger$ we have:

$$A^{MPL}(v) = \int_0^v x^{MPL}(s)ds = \mu v + \mu\int_0^v W(-e^{-1-s/\mu})ds.$$

Denoting $w = W(-e^{-1-s/\mu})$, we have $we^w = -e^{-1-s/\mu}$ by the definition of Lambert W function, and so,

$$s = -\mu\big(\ln(-w) + w + 1\big) \implies ds = -\mu\big(\frac{1}{w} + 1\big)dw.$$

Define

$$\begin{aligned} AW(v) \equiv \int_0^v W(-e^{-1-s/\mu})ds = -\mu \int_{-1}^{W(-e^{-1-v/\mu})} (1+w)dw \\ = -\mu\Big(W(-e^{-1-v/\mu}) + \frac{1}{2}W(-e^{-1-v/\mu})^2 + \frac{1}{2}\Big). \end{aligned}$$

Let us focus on the case where μ is small. Therefore, $W(-e^{-1-v/\mu}) \approx 0$, and the first-order approximation is:

$$AW(v^\dagger) \approx -\frac{\mu}{2} \implies A^{MPL}(v^\dagger) = \mu v^\dagger + \mu AW(v^\dagger) \approx \mu v^\dagger.$$

- Suppose $v_{max} < 2$ and denote $v^+ = \min\{v^\dagger, 1 - \mu\bar{v}\}$. Thus,

$$A^{DE}(v^+) = \mu v^+ \bar{v} < \mu v^+ \approx A^{MPL}(v^+)$$

  Moreover, for $v > v^+$ we know that b^{MPL} is above b^{DE} because b^{MPL} cuts b^{DE} at $\hat{v}$ close to 0, when μ is small. Therefore,

$$A^{DE}(v_{max}) < A^{MPL}(v_{max}).$$

- On the other hand, if $v_{max} > 2$, then $v^\dagger < 1 - \mu\bar{v}$, as we showed before. Let us ignore the part after $1 - \mu\bar{v}$, since both bids are c_{max}. Focusing on values up to that point,

$$\begin{aligned} A^{MPL}(1-\mu\bar{v}) &= A^{MPL}(v^\dagger) + \int_{v^\dagger}^{1-\mu\bar{v}} (1-s)ds \\ &= A^{MPL}(v^\dagger) + \frac{1}{2}\big((1-v^\dagger)^2 - (\mu\bar{v})^2\big) \\ &\approx \mu v^\dagger + \frac{1}{2}(1-v^\dagger)^2. \end{aligned}$$

  Furthermore,

$$A^{DE}(1-\mu\bar{v}) = \mu\bar{v}(1-\mu\bar{v}) \approx \mu\bar{v}.$$

  Therefore,

$$\begin{aligned} (A^{MPL} - A^{DE})(1-\mu\bar{v}) &\approx \mu v^\dagger + \frac{1}{2}(1-v^\dagger)^2 - \mu\bar{v} \\ &> \mu v^\dagger + \frac{1}{2}(2\bar{v} - \frac{3}{2})\mu - \mu\bar{v} = \mu(v^\dagger - \frac{3}{4}). \end{aligned}$$

The inequality holds for small enough μ, and comes from Lemma 1, using $k = 2\bar{v} - \frac{3}{2}$ and $\alpha = \frac{1}{2}$. Consequently, since $v^{\dagger}$ goes to 1 as μ goes to zero, for small enough μ we have:

$$A^{DE} < A^{MPL}.$$

Thus, the average inflation factor is always higher under *MPL* for μ small when the distributions of types and costs are both uniform. Note that we only considered first-order approximations with respect to μ and found that $A^{MPL} - A^{DE}$ is positive and of order μ.

D Decision Screens

Figure D.1: Decision Screen DE

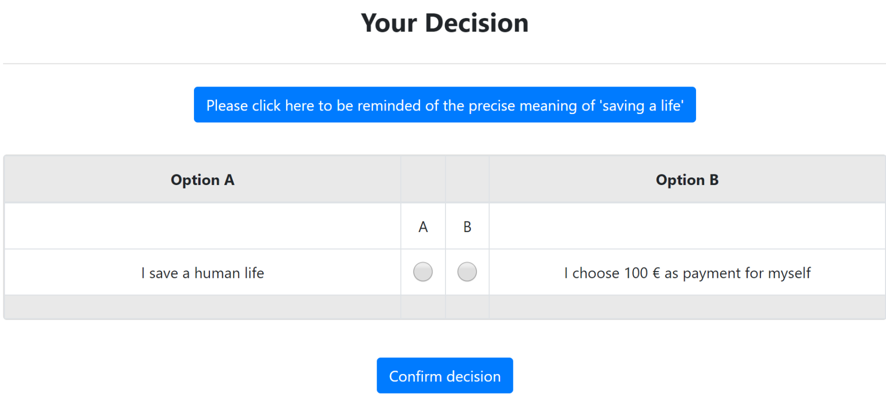
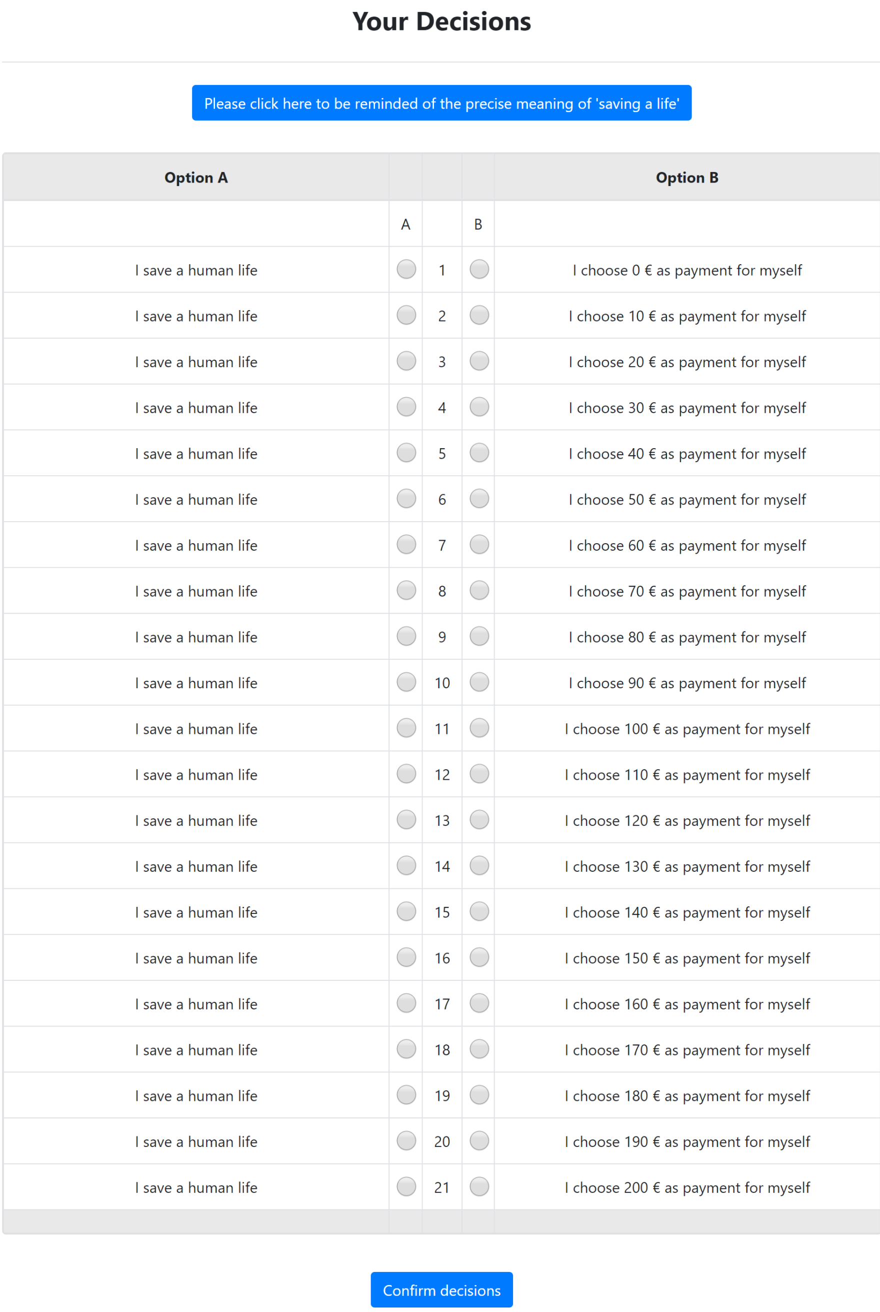


Figure D.2: Decision Screen MPL



Your Decisions

Please click here to be reminded of the precise meaning of 'saving a life'

Option A				Option B
	A		B	
I save a human life		1		I choose 0 € as payment for myself
I save a human life		2		I choose 10 € as payment for myself
I save a human life		3		I choose 20 € as payment for myself
I save a human life		4		I choose 30 € as payment for myself
I save a human life		5		I choose 40 € as payment for myself
I save a human life		6		I choose 50 € as payment for myself
I save a human life		7		I choose 60 € as payment for myself
I save a human life		8		I choose 70 € as payment for myself
I save a human life		9		I choose 80 € as payment for myself
I save a human life		10		I choose 90 € as payment for myself
I save a human life		11		I choose 100 € as payment for myself
I save a human life		12		I choose 110 € as payment for myself
I save a human life		13		I choose 120 € as payment for myself
I save a human life		14		I choose 130 € as payment for myself
I save a human life		15		I choose 140 € as payment for myself
I save a human life		16		I choose 150 € as payment for myself
I save a human life		17		I choose 160 € as payment for myself
I save a human life		18		I choose 170 € as payment for myself
I save a human life		19		I choose 180 € as payment for myself
I save a human life		20		I choose 190 € as payment for myself
I save a human life		21		I choose 200 € as payment for myself

Confirm decisions

E Instructions

E.1 Announcement by the Experimenter

The following text was read aloud by the experimenter after all subjects were placed in their cubicles, establishing common knowledge among all subjects of a session. The content depended on the image treatment.

E.1.1 Treatment *Low Image*

Welcome to today's study. In today's study, you will make decisions on a computer. These decisions will take place under complete anonymity. To ensure this, we will now apply the following procedure: You should all have two notes with your cubicle number in front of you. We will soon collect one of the two notes and randomly draw one out of all collected. The person in the drawn cubicle is responsible for the payment in today's study. At the end of the study, we prepare sealed envelopes with your payments. Those envelopes are then passed to the soon to be randomly drawn person, who will hand them out to each of you sequentially in the adjacent room. The envelopes are designed so that you cannot see the contents from the outside, i.e., not on weight or similar clues. Hence at no time can there be a connection drawn between your payment and your decisions. Please hold now one of the notes with your cubicle number onto out of your cubicle. (Responsible person is drawn and placed in the adjacent room) The study will begin shortly. If you have at any time have questions, just hold your hand out of the cubicle.

E.1.2 Treatment *High Image*

Welcome to today's study. In today's study, you will make decisions on your computer. Your decisions will subsequently be evaluated by a committee consisting of three students from the University of Bonn. For this, after you have made your decisions, you will go to the adjacent room, where your decisions will be projected on a wall with a projector. You will then briefly communicate your decisions to the committee, and the committee will evaluate them. Afterward, you will receive the result of the evaluation. Detailed information about your decisions, the committee, and the evaluation will be given to you at the appropriate time on your computer. The study will begin shortly. If you have at any time have questions, just hold your hand out of the cubicle.

E.1.3 Further Procedure

After the text was read aloud, in the *Low Image* conditions the experimenter then collected one note from each subject indicating their respective cabin number. All notes were thrown into a bag, and one was drawn in front of all participants to make clear that the person responsible for the payment procedure was a randomly determined participant. In the *High Image* conditions, subjects were shown the adjacent room and the setup with the committee, which consisted of student research assistants. The members of the committee did not interact with the subjects in any way.

E.2 Introduction

All further instructions were displayed on the subjects screens. The following introduction was the same for all treatments.

E.2.1 Welcome to the study

Welcome, and thank you for your interest in today's study!
For your participation, you will receive a fixed payment of 12€ given to you at the end. In this study, you will make decisions on the computer. Depending on how you choose, you can earn additional money.
During the entire study, communication between participants is prohibited. Please turn off your phone so that other participants are not disturbed. Please only use the designated functions on the computer and make the entries with the mouse and keyboard. If you, at some point, have questions, please make a hand signal. Your question will be answered at your seat.
On the next screens, you will receive specific information about participation in this study. To proceed, click "Next".

E.2.2 Your Partner

As part of this experiment, a partner has been assigned to you. This partner is a participant in today's experiment, just like you. He or she was randomly assigned to you and will receive the same instructions as you.
In today's experiment, you and your partner will both receive the exact same information and subsequently face the exact same decisions. These decisions have certain consequences, which will be described in detail later.
At the end of today's experiment, one pair is randomly drawn from all participants in today's experiment. Only the decisions of this pair will be implemented, as described in the instructions. Please note: The random draw of a pair is completely independent of the participants' decisions. Each pair has the same probability of being drawn. Since your decision can be actually implemented for real, you should think carefully about how you will decide in the experiment.

Figure E.1: Typical appearance of a tuberculosis patient.

E.2.3 Information about Tuberculosis

What follows is important information that is relevant to the decisions you will later be asked to make. It concerns the illness tuberculosis and its possible treatment. Please read through all the information carefully.

What is Tuberculosis?

Tuberculosis – also called Phthisis or White Death – is an infectious disease, which is caused by bacteria. Roughly one-third of all humans are infected with the pathogen of Tuberculosis. Active Tuberculosis breaks out among 5 to 10% of all those infected. Tuberculosis is primarily airborne. This is also why quick treatment is necessary.

Tuberculosis patients often suffer from very unspecific symptoms like fatigue, the feeling of weakness, lack of appetite, and weight loss. At an advanced stage of lung tuberculosis, the patient coughs up blood, leading to the so-called rush of blood. Without treatment, a person with Tuberculosis dies with a probability of 43%.

How prevalent is Tuberculosis?

In the year 2014, 6 million people have been recorded as falling ill with active Tuberculosis. Almost 1.5 million people die of Tuberculosis each year. This means more deaths due to Tuberculosis than due to HIV, malaria, or any other infectious disease.

Is tuberculosis curable?

According to the World Health Organization (WHO), the United Nations agency for international public health, "tuberculosis is preventable and curable". Treatment takes place by taking antibiotics several times a week over a period of 6 months. It is important to take the medication consistently. Since 2000, an estimated 53 million lives have been saved through effective diagnosis and treatment of tuberculosis.

The success rate of treatment for a new infection is usually over 85%.

The preceding figures and information have been provided by the WHO and are freely available. Click here for more details.

Figure E.2: A worker from Operation ASHA delivers medication to a tuberculosis patient.

Operation ASHA

Operation ASHA is a charity organization specialized since 2005 on treating Tuberculosis in disadvantaged communities. The work of *Operation ASHA* is based on the insight that the biggest obstacle for the treatment of Tuberculosis is the interruption of the necessary 6-month-long regular intake of medication.

For a successful treatment, the patient has to come to a medical facility twice a week – more than 60 times in total – to take the medication. Interruption or termination of the treatment is fatal because this strongly enhances the development of a drug-resistant form of Tuberculosis. This form of Tuberculosis is much more difficult to treat and almost always leads to death.

The Concept of *Operation ASHA*

To overcome this problem, *Operation ASHA* developed a concept that guarantees regular treatment through immediate spatial proximity to the patient. A possible non-adherence is additionally prevented by visiting the patient at home.

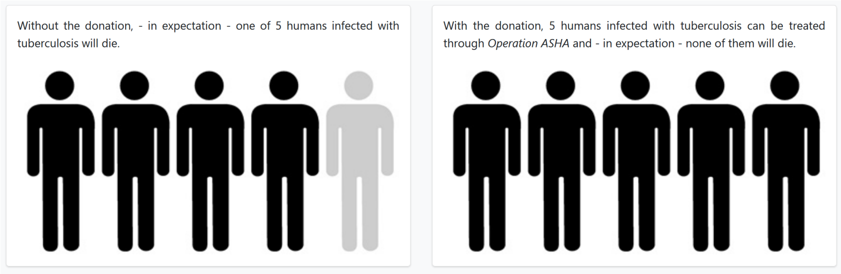
By now, *Operation ASHA* runs more than 360 treatment centers, almost all of which are located in the poorer regions of India. More than 60,000 sick persons have been identified and treated that way.

Operation ASHA is an internationally recognized organization, and its success has been covered by the New York Times, BBC, and Deutsche Welle, for example. The MIT and the University College London have already conducted research projects about the fight against Tuberculosis in cooperation with *Operation ASHA*. The treatment method employed by *Operation ASHA* is described by the World Health Organization (WHO) as "highly efficient and cost-effective".

The Impact of a Donation to *Operation ASHA*

It is now possible to save people from death by Tuberculosis by donating to *Operation ASHA*.

Figure E.3: Relationship between the donation and the saving of a life



To save a person's life means here to successfully cure a person with Tuberculosis, who otherwise would die because of the Tuberculosis. A donation of 350€ ensures that at least one human life can be expected to be saved. The information used to calculate the donation amount is obtained from public statements from the World Health Organization (WHO), peer-reviewed research studies, Indian Government statistics, and published figures from *Operation ASHA*.

In the calculation, information was conservatively interpreted, or a pessimistic number was used so that the donation amount of 350€ is in the case of doubt higher than the actual costs to save a human life. In addition, in the calculation of the treatment success rate of *Operation ASHA*, the mortality rate for alternative treatment by the state tuberculosis program in India and the different detection rates for new cases of Tuberculosis are included.

In the context of this study, an agreement made with *Operation ASHA* will ensure that 100% of the donation will be used exclusively for the diagnosis and treatment of tuberculosis patients. This means that every Euro of the donation amount goes directly to saving human lives, and no other costs will be covered. Based on a very high number of cases, the contribution of a donation of 350€ can be simplified visualized as follows:

With a donation of 350€ 5 additional patients infected with Tuberculosis can be treated through *Operation ASHA*.

If these 5 persons are not treated through *Operation ASHA*, it is expected that one patient will die.

If, through the donation of 350€ all 5 patients are treated, it is expected that no patient will die.

Based on this experience, this means that through a donation of 350€ the life of a human will be saved. The relationship between a donation of 350€ and the saving of a human is illustrated in the following graphic: [Figure E.3 here]

Summary

Tuberculosis is a worldwide common bacterial infectious disease. The success rate of medical treatment of a new disease is very high. Nevertheless, close to 1.5 million people die every year from Tuberculosis. The biggest obstacle to the curing of Tuberculosis is the potential stopping of continuous treatment with antibiotics. The concept of *Operation ASHA* is therefore based on the immediate proximity to the patient as well as the control and recording of the regular intake of medication. Through a donation of 350€ to *Operation ASHA*, a life will be saved.

How is the donation connected to the saving of a life?

The donation of 350€ already accounts for the fact that someone inflicted with the illness could have survived without treatment by *Operation ASHA*; i.e., instead of through *Operation ASHA*, they could have received treatment through other actors (such as the public health system). The amount is, therefore, sufficient for the diagnosis and complete treatment of multiple sufferers.

What does it mean to "save a life"?

To save a life means here the successful curing of a person suffering from Tuberculosis, who otherwise would die because of Tuberculosis. In particular, this means that the amount of the donation is sufficient to identify and cure so many tuberculosis patients that there is at least one person among them who otherwise could be anticipated to have died of Tuberculosis.

Note

Click on "Next" once you have finished carefully reading through the information.

You can only click on the button "Next" once you have spent at least 5 minutes on the tabs of this page.

E.3 Treatment *DE Low Image*

E.3.1 Your Decision

You will soon have the possibility to choose between two options: option A and option B. Both options are as follows:

Option A

Option A: I save a human life. By choosing option A, you save a human life. Specifically, by choosing option A, you instigate a donation of 350,00€ that will ensure that at least one person is saved from death by Tuberculosis, just as described before. If you choose option A, you will not receive an additional payment.

Option B

Option B: I choose X€ as payment for myself. By choosing option B, you will receive an additional payment at the end of the experiment. In addition, the absence of your donation will cause the death of a human life.

Additional Payment

Before today's experiment, various amounts between 0€ and 200€ were taken into account for the amount of money you will receive when choosing option B, from which 100€ was selected. Your partner sees exactly the same options as you and makes a decision just like you. So your partner also decides between option A (saving a human life) and option B (keeping 100€ to himself).

Summary

You will decide on the next page of the screen by choosing between option A and option B. By choosing option A, you save a human life. By choosing option B, you receive an additional payment of 100€. On the next page, you will receive details about the payment procedure.

E.3.2 Further Procedure

After you confirmed your decision on the decision screen, a screenshot will be taken from this decision screen. From the decision screen of your partner, a screenshot will be taken in the same way. Thereafter, some additional questions will follow. After you have answered these questions, you will get the screenshot with the decision of your partner displayed, and your partner will get the screenshot with your decision. You will not receive any further information about your partner, and your partner will not receive any further information about you.

After you received the screenshot, please remain seated until you are called with your cabin number. Then you can go into the adjacent room to pick up your compensation for today's experiment. You will be called one by one so that there is no contact with other participants of the experiment.

Who will be in the adjacent room?

In the adjacent room, you will find the participant who was randomly selected from all participants at the start of the study.

How do you receive your payment?

This participant will give you a sealed envelope with your payment. The selected participant has already received the envelope sealed. Since this participant is only responsible for the payment, this participant has not completed the study and therefore has no knowledge of the decisions to be made. Therefore, this participant does not know what you chose, how you decided, or how much money you received, exactly as explained at the beginning of the study. By handing in your note with your cabin number, you will receive the envelope intended for you.

Data protection

The subsequent analysis of all data is carried out anonymously so that your decision can never be linked to your person. Your anonymity is therefore always guaranteed, and the information about your decision is only used for anonymized data analysis.

Please note:

This is not a thought experiment: All information given in these instructions is true. In particular, all actions are performed exactly as they are described. This fundamentally applies to all studies of the Bonn Laboratory for Experimental Economic Research, as well as to this study.

If you still have separate questions, you may send them to experimente@briq-institute.org after the study.

E.3.3 Comprehension Questions

Question 1: For this question, hypothetically assume that you made your decisions as follows: [Example of DE decision with donation selected]
Question 1a: How many euros would you receive as an additional payment in this case?
Question 1b: Would you have saved a human life?
Question 1c: If, in this decision situation, you had chosen option B instead of option A, how many euros would you have received as an additional payment?
Question 1d: Would you have saved a human life in this case?
Question 2: For this question, hypothetically assume that you made your decisions as shown in the table on the left, while your partner made their decisions as shown in the table on the right. [Example of DE decision with the donation selected on the left, and the monetary payment selected on the right table] What are the consequences for you and your partner?

E.4 Treatment *DE High Image*

E.4.1 Your Decision

You will soon have the possibility to choose between two options: option A and option B. Both options are as follows:
Option A
Option A: I save a human life. By choosing option A, you save a human life. Specifically, by choosing option A, you instigate a donation of 350,00€ that will ensure that at least one person is saved from death by Tuberculosis, just as described before. If you choose option A, you will not receive an additional payment.
Option B
Option B: I choose X€ as payment for myself. By choosing option B, you will receive an additional payment at the end of the experiment. In addition, the absence of your donation will cause the death of a human life.
Additional Payment
Before today's experiment, various amounts between 0€ and 200€ were taken into account for the amount of money you will receive when choosing option B, from which 100€ was selected. Your partner sees exactly the same options as you and makes a decision just like you. So your partner also decides between option A (saving a human life) and option B (keeping 100€ to himself).
Summary
You will decide on the next page of the screen by choosing between option A and option B. By choosing option A, you save a human life. By choosing option B, you receive an additional payment of 100€. On the next page, you will receive details about the payment procedure.

E.4.2 Further Procedure

After you confirmed your decision on the decision screen, a screenshot will be taken from this decision screen. From the decision screen of your partner, a screenshot will be taken in the same way. Thereafter, some additional questions will follow. After you have answered these questions, you will get the screenshot with the decision of your partner displayed, and your partner will get the screenshot with your decision. You will not receive any further information about your partner, and your partner will not receive any further information about you.

After you received the screenshot, please remain seated until you are called with your cabin number. Then you can go into the adjacent room to pick up your compensation for today's experiment. You will be called one by one so that there is no contact with other participants of the experiment.

Who will be in the adjacent room?

In the adjacent room, you will find a person who will make the payment. As mentioned before, there will also be a committee consisting of three other persons in the adjacent room. These three persons are students of the University of Bonn and were specially selected for this task.

What is the task of the committee?

The task of these three persons is to assess the decision you and your partner have taken. Specifically, the assessment is about how moral your behavior and the behavior of your partner is. Apart from the assessment, the three persons will not interact with you (or with your partner) in any way, and the rating will not influence at all the consequences of your decisions or your payment.

What information does the committee receive?

In order for the committee to make the assessment, the two screenshots of the decision of you and your partner are projected side by side onto the room's wall using a projector, visibly for all persons in the adjacent room. You are identified by (and only by) your cabin number. For better identification, based on your decision and that of your partner, you must also say the following two sentences aloud. The first sentence refers to your decision, the second sentence to your partner's decision.

Sentence 1 in case you chose option A: "I have decided not to take 100€ as payment for myself and instead decided to save a human life."Sentence 1 in case you chose option B: "I have decided to take 100€ as payment for myself instead of saving a human life."

Sentence 2 in case your partner has chosen option A: "My partner has decided not to take 100€ as payment for himself and instead decided to save a human life."Sentence 2 in case your partner has chosen option A: "My partner has decided to take 100€ as payment for himself instead of saving a human life."

In summary, you have to say two sentences, and the following information will be visible to everyone in the room:

- The decision you and your partner faced.
- Which option you and your partner have chosen. This means it is displayed whether you chose to save a human life or the additional payment of 100€ and whether your partner chose to save a human life or the additional payment of 100€.

How does the assessment work?
The committee will assess your decision using a scale. For this, each one of the three persons of the committee selects one of the following five values:
1 - very immoral 2 - rather immoral 3 - neutral 4 - rather moral 5 - very moral.
The three persons of the committee will submit an assessment for your decision as well as the decision of your partner.
How do you receive your payment?
After the committee has assessed the decisions, the committee will give you the assessments of both your decision and the decision of your partner, and the person responsible for the payments will give you your payment. In the event that you have decided to donate, you will receive a donation confirmation.
Data protection
The subsequent analysis of all data is carried out anonymously so that your decision can never be linked to your person. Your anonymity is therefore always guaranteed, and the information about your decision is only used for anonymized data analysis.
Please note:
This is not a thought experiment: All information given in these instructions is true. In particular, all actions are performed exactly as they are described. This fundamentally applies to all studies of the Bonn Laboratory for Experimental Economic Research, as well as to this study.
If you still have separate questions, you may send them to experimente@briq-institute.org after the study.

E.4.3 Comprehension Questions

Question 1: For this question, hypothetically assume that you made your decisions as follows:
[Example of DE decision with donation selected]
Question 1a: How many euros would you receive as an additional payment in this case?
Question 1b: Would you have saved a human life?
Question 1c: If, in this decision situation, you had chosen option B instead of option A, how many euros would you have received as an additional payment?
Question 1d: Would you have saved a human life in this case?

Question 2: For this question, hypothetically assume that you made your decisions as shown in the table on the left, while your partner made their decisions as shown in the table on the right. [Example of DE decision with the donation selected on the left, and the monetary payment selected on the right table]
Question 2a: What are the consequences for you and your partner?
Question 2b: Which two sentences would you have to say in front of the committee in the adjacent room in this case?
Question 3: How many people are in the committee that will evaluate your and your partner's decisions in the adjacent room?

E.5 Treatment *MPL Low Image*

E.5.1 Your Decision

You will soon have the possibility to choose in 21 decision scenarios between two options: option A and option B. Both options are as follows:
Option A
Option A: I save a human life. By choosing option A, you save a human life. Specifically, by choosing option A, you instigate a donation of 350,00€ that will ensure that at least one person is saved from death by Tuberculosis, just as described before. If you choose option A, you will not receive an additional payment.
Option B
Option B: I choose X€ as payment for myself. By choosing option B, you will receive an additional payment at the end of the experiment. In addition, the absence of your donation will cause the death of a human life.
Additional Payment
The additional payment that you receive from choosing option B varies in each of the 21 decision scenarios. In the first scenario, the payment is 0€ and then increases incrementally in each scenario thereafter by 10€ up to a payment of 200€. Therefore, the decision scenarios look as follows:
Automatic Completion Help
So that you do not need to click as much, we have activated an automatic completion help that automatically fills out the fields for you. As soon as you choose an amount from option B, we assume that you would choose all respectively higher payments from option B. Likewise, when you choose option A in a row, we assume that you would choose option A over all respectively lower payments from option B.
Please note: You can always change your decisions until you clicked on "Confirm Decisions". Therefore, only click on that button when you are certain how you want to decide.
Payment

After you have selected one of the two options for each of the 21 decision scenarios, one of them will be randomly selected for real implementation. This means that the consequences of this decision will be implemented exactly as stated. Each of the 21 scenarios has the same probability of being selected. Therefore, since each of your decisions is potentially relevant, it is in your interest to decide in every scenario as if that decision is being implemented for real.

Your partner sees exactly the same 21 decision scenarios as you and, like you, makes a decision for every scenario. Furthermore, for you and your partner, the same decision scenario will be randomly selected. Thus, both your decision and the decision of your partner for this scenario will be implemented.

The following examples elaborate on this. Assume that decision scenario 2 is randomly selected, and you chose option A, while your partner chose option B. Then you save a human life and your partner will receive 10€. If, on the contrary, both of you choose option B, then both of you will receive 10€. If both of you choose option A, then two human lives will be saved. Assuming that decision scenario 21 is randomly selected, and you chose option B, while your partner chose option A. Then, you will receive 200€ and your partner saves a human life. If, however, both of you chose option B, then both of you will receive 200€. If both of you chose option A, then two human lives will be saved.

Summary

On the page after next, you will make a decision for 21 scenarios, and in each decision, you can choose between option A and option B. By choosing option A, you save a human life, whereas by choosing option B, you receive an additional payment. After you have reached all of your decisions, one of the 21 scenarios will be chosen randomly for you and your assigned partner. Thereafter, the consequences of the chosen decision are realized, i.e., in the case that you chose option A under this scenario, a donation will be made towards the saving of a human life and in the case that you chose option B, you receive the respective amount from the selected scenario. The same applies to your partner. On the next page, you will receive details about the payment procedure.

E.5.2 Further Procedure

After you confirmed your decisions on the decision screen, a screenshot will be taken from this decision screen. From the decision screen of your partner, a screenshot will be taken in the same way. Thereafter, some additional questions will follow. After you have answered these questions, you will get the screenshot with the decisions of your partner displayed, and your partner will get the screenshot with your decisions. You will not receive any further information about your partner, and your partner will not receive any further information about you.

After you received the screenshot, please remain seated until you are called with your cabin number. Then you can go into the adjacent room to pick up your compensation for today's experiment. You will be called one by one so that there is no contact with other participants of the experiment.

Who will be in the adjacent room?

In the adjacent room, you will find the participant who was randomly selected from all participants at the start of the study.

How do you receive your payment?

This participant will give you a sealed envelope with your payment. The selected participant has already received the envelope sealed. Since this participant is only responsible for the payment, this participant has not completed the study and therefore has no knowledge of the decisions to be made. Therefore, this participant does not know what you chose, how you decided, or how much money you received, exactly as explained at the beginning of the study. By handing in your note with your cabin number, you will receive the envelope intended for you.

Data protection

The subsequent analysis of all data is carried out anonymously so that your decisions can never be linked to your person. Your anonymity is therefore always guaranteed, and the information about your decisions is only used for anonymized data analysis.

Please note:

This is not a thought experiment: All information given in these instructions is true. In particular, all actions are performed exactly as they are described. This fundamentally applies to all studies of the Bonn Laboratory for Experimental Economic Research, as well as to this study.

If you still have separate questions, you may send them to experimente@briq-institute.org after the study.

E.5.3 Comprehension Questions

Question 1: For this question, hypothetically assume that you made your decisions as follows: [Example of MPL decision with switching point at 160€] Also assume that the decision situation number 14 (highlighted in gray) [number 14 is 130€] was randomly selected for the real payout for you and your partner.

Question 1a: How many euros would you receive as an additional payment in this case?

Question 1b: Would you have saved a human life?

Question 1c: If, in this decision situation, you had chosen option B instead of option A, how many euros would you have received as an additional payment?

Question 1d: Would you have saved a human life in this case?

Question 2: For this question, hypothetically assume that you made your decisions as follows: [Example of MPL decision with switching point at 80€] Also assume that the decision

situation number 10 (highlighted in gray) [number 10 is 90€ was randomly selected for the real payout for you and your partner.
Question 2a: How many euros would you receive as an additional payment in this case?
Question 2b: Would you have saved a human life?
Question 2c: If, in this decision situation, you had chosen option A instead of option B, how many euros would you have received as an additional payment?
Question 2d: Would you have saved a human life in this case?
Question 3: For this question, hypothetically assume that you made your decisions as shown in the table on the left, while your partner made their decisions as shown in the table on the right. [Example of MPL decision with switching point at 18€for the left, and 200 for the right table] Also assume that the decision situation number 3 (highlighted in gray) [number 3 is 20€] was randomly selected for the real payout for you and your partner. What are the consequences for you and your partner?

E.6 Treatment *MPL High Image*

E.6.1 Your Decision

You will soon have the possibility to choose in 21 decision scenarios between two options: option A and option B. Both options are as follows:
Option A
Option A: I save a human life. By choosing option A, you save a human life. Specifically, by choosing option A, you instigate a donation of 350,00€ that will ensure that at least one person is saved from death by Tuberculosis, just as described before. If you choose option A, you will not receive an additional payment.
Option B
Option B: I choose X€ as payment for myself. By choosing option B, you will receive an additional payment at the end of the experiment. In addition, the absence of your donation will cause the death of a human life.
Additional Payment
The additional payment that you receive from choosing option B varies in each of the 21 decision scenarios. In the first scenario, the payment is 0€ and then increases incrementally in each scenario thereafter by 10€ up to a payment of 200€. Therefore, the decision scenarios look as follows:
Automatic Completion Help
So that you do not need to click as much, we have activated an automatic completion help that automatically fills out the fields for you. As soon as you choose an amount from option B, we assume that you would choose all respectively higher payments from option B. Likewise, when you choose option A in a row, we assume that you would choose option A over all respectively lower payments from option B.

Please note: You can always change your decisions until you clicked on “Confirm Decisions”. Therefore, only click on that button when you are certain how you want to decide.

Payment

After you have selected one of the two options for each of the 21 decision scenarios, one of them will be randomly selected for real implementation. This means that the consequences of this decision will be implemented exactly as stated. Each of the 21 scenarios has the same probability of being selected. Therefore, since each of your decisions is potentially relevant, it is in your interest to decide in every scenario as if that decision is being implemented for real.

Your partner sees exactly the same 21 decision scenarios as you and, like you, makes a decision for every scenario. Furthermore, for you and your partner, the same decision scenario will be randomly selected. Thus, both your decision and the decision of your partner for this scenario will be implemented.

The following examples elaborate on this. Assume that decision scenario 2 is randomly selected, and you chose option A, while your partner chose option B. Then you save a human life and your partner will receive 10€. If, on the contrary, both of you choose option B, then both of you will receive 10€. If both of you choose option A, then two human lives will be saved. Assuming that decision scenario 21 is randomly selected, and you chose option B, while your partner chose option A. Then, you will receive 200€ and your partner saves a human life. If, however, both of you chose option B, then both of you will receive 200€. If both of you chose option A, then two human lives will be saved.

Summary

On the page after next, you will make a decision for 21 scenarios, and in each decision, you can choose between option A and option B. By choosing option A, you save a human life, whereas by choosing option B, you receive an additional payment. After you have reached all of your decisions, one of the 21 scenarios will be chosen randomly for you and your assigned partner. Thereafter, the consequences of the chosen decision are realized, i.e., in the case that you chose option A under this scenario, a donation will be made towards the saving of a human life and in the case that you chose option B, you receive the respective amount from the selected scenario. The same applies to your partner. On the next page, you will receive details about the payment procedure.

E.6.2 Further Procedure

After you confirmed your decisions on the decision screen, a screenshot will be taken from this decision screen. From the decision screen of your partner, a screenshot will be taken in the same way. Thereafter, some additional questions will follow. After you have answered these questions, you will get the screenshot with the decisions of your partner displayed, and your partner will get the screenshot with your decisions. You will not receive any further

information about your partner, and your partner will not receive any further information about you.
After you received the screenshot, please remain seated until you are called with your cabin number. Then you can go into the adjacent room to pick up your compensation for today's experiment. You will be called one by one so that there is no contact with other participants of the experiment.

Who will be in the adjacent room?

In the adjacent room, you will find a person who will make the payment. As mentioned before, there will also be a committee consisting of three other persons in the adjacent room. These three persons are students of the University of Bonn and were specially selected for this task.

What is the task of the committee?

The task of these three persons is to assess the decisions you and your partner have taken. Specifically, the assessment is about how moral your behavior and the behavior of your partner is. Apart from the assessment, the three persons will not interact with you (or with your partner) in any way, and the rating will not influence at all the consequences of your decisions or your payment.

What information does the committee receive?

In order for the committee to make the assessment, the two screenshots of the decisions of you and your partner are projected side by side onto the room's wall using a projector, visibly for all persons in the adjacent room. You are identified by (and only by) your cabin number. For better identification, based on your decisions and the decisions of your partner, you must also say the following two sentences aloud. The first sentence refers to your decisions, the second sentence to your partner's decisions.
Sentence 1: "I have decided from a payment of X€ onwards to take the payment for myself instead of saving human life."
Sentence 2: "My partner has decided from a payment of X€ onwards to take the payment for himself instead of saving human life."
The payment X denotes the amount of money for which you switched from option A to option B for the first time. If you have not decided to take the money in any decision-making situation, i.e., have not switched, you have to say the following as the first sentence:
Sentence 1: "I have decided for no amount to take the payment for myself instead of saving human life."
Similarly, if your partner has not decided to take the money in any decision-making situation, you must say the following second sentence:
Sentence 2: "My partner has decided for no amount to take the payment for himself instead of saving human life."
In summary, you have to say two sentences, and the following information will be visible to everyone in the room:

- The complete list of all 21 decision scenarios described before.
- How you and your partner have chosen in each of these scenarios. This means that for each payment amount, one can see whether you have decided to save a human life or the additional payment and whether your partner has decided to save a human life or the additional payment.

How does the assessment work?
The committee will assess your decisions using a scale. For this, each one of the three persons of the committee selects one of the following five values:
1 - very immoral 2 - rather immoral 3 - neutral 4 - rather moral 5 - very moral.
The three persons of the committee will submit an assessment for your decisions as well as the decisions of your partner.
How do you receive your payment?
After the committee has assessed the decisions, the committee will give you the assessments of both your decisions and the decisions of your partner, and the person responsible for the payments will give you your payment. In the event that you have decided to donate, you will receive a donation confirmation.
Data protection
The subsequent analysis of all data is carried out anonymously so that your decisions can never be linked to your person. Your anonymity is therefore always guaranteed, and the information about your decisions is only used for anonymized data analysis.
Please note:
This is not a thought experiment: All information given in these instructions is true. In particular, all actions are performed exactly as they are described. This fundamentally applies to all studies of the Bonn Laboratory for Experimental Economic Research, as well as to this study.
If you still have separate questions, you may send them to experimente@briq-institute.org after the study.

E.6.3 Comprehension Questions

Question 1: For this question, hypothetically assume that you made your decisions as follows: [Example of MPL decision with switching point at 160€] Also assume that the decision situation number 14 (highlighted in gray) [number 14 is 130€] was randomly selected for the real payout for you and your partner.
Question 1a: How many euros would you receive as an additional payment in this case?
Question 1b: Would you have saved a human life?
Question 1c: If, in this decision situation, you had chosen option B instead of option A, how many euros would you have received as an additional payment?

Question 1d: Would you have saved a human life in this case?
Question 2: For this question, hypothetically assume that you made your decisions as follows: [Example of MPL decision with switching point at 80€] Also assume that the decision situation number 10 (highlighted in gray) [number 10 is 90€ was randomly selected for the real payout for you and your partner.
Question 2a: How many euros would you receive as an additional payment in this case?
Question 2b: Would you have saved a human life?
Question 2c: If, in this decision situation, you had chosen option A instead of option B, how many euros would you have received as an additional payment?
Question 2d: Would you have saved a human life in this case?
Question 3: For this question, hypothetically assume that you made your decisions as shown in the table on the left, while your partner made their decisions as shown in the table on the right. [Example of MPL decision with switching point at 18€ for the left, and 200 for the right table] Also assume that the decision situation number 3 (highlighted in gray) [number 3 is 20€] was randomly selected for the real payout for you and your partner.
Question 3a: Which two sentences would you have to say in front of the committee in the adjacent room in this case?
Question 3b: What are the consequences for you and your partner?
Question 4: How many people are in the committee that will evaluate your and your partner's decisions in the adjacent room?

E.7 Robustness Experiment

E.7.1 Introduction

All instructions were displayed on the subjects' screens. The following introduction was the same for both treatments of the robustness experiment.

E.7.2 Welcome to the study

Welcome, and thank you for your interest in today's study!
Please note that you can take part in this study only once. Furthermore, you may only participate if you have registered for this study in our participation database (`experimente.bonneconlab.uni-bonn.de`).
For your full participation, you will receive a fixed payment of 3€. In this study, you will make decisions on the computer. Depending on how you choose, you can earn additional money. After the study, you will receive all payments, i.e. both the remuneration for your participation and any additional payments based on your decisions, by bank transfer.
On the next screens, you will receive specific information about participation in this study. To proceed, click "Next".

E.7.3 Your Partner

As part of this experiment, a partner has been assigned to you. This partner is a participant in today's experiment, just like you. He or she was randomly assigned to you and will receive the same instructions as you.
In today's experiment, you and your partner will both receive the exact same information and subsequently face the exact same decisions. These decisions have certain consequences, which will be described in detail later.

Payment

At the end of today's experiment, one pair will be randomly drawn from every 24 participants in the experiment. Only the decisions of this pair will be implemented, as described in the instructions. Please note: The random draw of a pair is completely independent of the participants' decisions. Each pair has the same probability of being drawn. Since your decision can be actually implemented for real, you should think carefully about how you will decide in the experiment.

E.7.4 Information

What follows is some information that is relevant to the decisions you will later be asked to make. It concerns the official shop of the University of Bonn.
The Campus Store Uni-Bonn is the official shop of the University of Bonn. Here you can purchase various products such as T-shirts, sweatshirts or mugs with the logo and design of the Uni-Bonn.
The Uni-shop is located at the information point in the main building. There is also an online shop, which can be reached via the website: `https://www.campusstore-unibonn.de`.
The online shop dispatches all goods within 2-3 working days.

Voucher

The next decisions will concern a voucher for the Uni-shop, namely a voucher worth 35€. The voucher can only be redeemed in the online shop and cannot be converted into money.

E.8 Treatment *DE No-Image*

E.8.1 Your Decision

You will soon have the possibility to choose between two options: option A and option B. Both options are as follows:

Option A

Option A: I choose the voucher. By choosing option A, you will receive the voucher for the Uni-shop. Specifically, option A allows you to receive a voucher worth 35€, which you can redeem in the Uni-shop (and only there). If you choose option A, you will not receive an additional payment.

Option B
Option B: I choose 10€ as payment for myself. By choosing option B, you will receive an additional payment of 10€ at the end of the experiment, but you will not receive the voucher.
Additional Payment
Before today's experiment, various amounts between 0€ and 20€ were taken into account for the amount of money you will receive when choosing option B, from which 10€ was selected. Your partner sees exactly the same options as you and makes a decision just like you. So your partner also decides between option A (voucher) and option B (keeping 10€ to himself/herself).
Summary
You will decide on the next page of the screen by choosing between option A and option B. By choosing option A, you receive a voucher. By choosing option B, you receive an additional payment of 10€. On the next page, you will find details about the payment procedure.

E.8.2 Further Procedure

After you confirmed your decision on the decision screen, a screenshot will be taken from this decision screen. From the decision screen of your partner, a screenshot will be taken in the same way. At the end of today's experiment, you will get the screenshot with the decision of your partner displayed, and your partner will get the screenshot with your decision. You will not receive any further information about your partner, and your partner will not receive any further information about you.
Data protection
The subsequent analysis of all data is carried out anonymously so that your decision can never be linked to your person. Your anonymity is therefore always guaranteed, and the information about your decision is only used for anonymized data analysis.
Please note:
This is not a thought experiment: All information given in these instructions is true. In particular, all actions are performed exactly as they are described. This fundamentally applies to all studies of the Bonn Laboratory for Experimental Economic Research, as well as to this study.
If you still have separate questions, you may send them to experiment@briq-institute.org after the study.

E.8.3 Comprehension Questions

Question 1 For this question, hypothetically assume that you made your decisions as follows:
[Example of DE decision with voucher selected]
Question 1a: How many euros would you receive as an additional payment (not in the form of the voucher) in this case?

Question 1b: Would you have received the voucher?
Question 1c: If, in this decision situation, you had chosen option B instead of option A, how many euros would you have received as an additional payment?
Question 1d: Would you have received the voucher in this case?
Question 2: For this question, hypothetically assume that you made your decisions as shown in the table on the left, while your partner made their decisions as shown in the table on the right. [Example of DE decision with the voucher selected on the left, and the monetary payment selected on the right table] What are the consequences for you and your partner?

E.9 Treatment *MPL No-Image*

E.9.1 Your Decisions

You will soon have the possibility to choose in 21 decision scenarios between two options: option A and option B. Both options are as follows:

Option A

Option A: I choose the voucher. By choosing option A, you will receive the voucher for the Uni-shop. Specifically, option A allows you to receive a voucher worth 35€, which you can redeem in the Uni-shop (and only there). If you choose option A, you will not receive an additional payment.

Option B

Option B: I choose X€ as payment for myself. By choosing option B, you will receive an additional payment at the end of the experiment, but you will not receive the voucher.

Additional Payment

The additional payment that you receive from choosing option B varies in each of the 21 decision scenarios. In the first scenario, the payment is 0€ and then increases incrementally in each scenario thereafter by 1€, up to a payment of 20€. Therefore, the decision scenarios look as follows:

Automatic Completion Help

So that you do not need to click as much, we have activated an automatic completion help that automatically fills out the fields for you. As soon as you choose an amount from option B, we assume that you would choose all respectively higher payments from option B. Likewise, when you choose option A in a row, we assume that you would choose option A over all respectively lower payments from option B.

Please note: You can always change your decisions until you clicked on "Confirm Decisions". Therefore, click on that button only when you are certain how you want to decide.

Payment

After you have selected one of the two options for each of the 21 decision scenarios, one of them will be randomly selected for real implementation. This means that the consequences of this decision will be implemented exactly as stated. Each of the 21 scenarios has the same probability of being selected. Therefore, since each of your decisions is potentially relevant, it is in your interest to decide in every scenario as if that decision is being implemented for real.

Your partner sees exactly the same 21 decision scenarios as you and, like you, makes a decision for every scenario. Furthermore, for you and your partner, the same decision scenario will be randomly selected. Thus, both your decision and the decision of your partner for this scenario will be implemented.

The following examples elaborate on this. Assume that decision scenario 2 is randomly selected, and you chose option A, while your partner chose option B. Then you will receive the voucher and your partner will receive 1€. If, on the contrary, both of you chose option B, then both of you will receive 1€. If both of you chose option A, then you and your partner will each receive the voucher. Assuming that decision scenario 21 is randomly selected, and you chose option B while your partner chose option A, then you will receive 20€, and your partner will receive the voucher. If, however, both of you chose option B, then both of you will receive 20€. If both of you chose option A, then you and your partner will each receive the voucher, etc.

Summary

On the page after next, you will make a decision for 21 scenarios, and in each decision, you can choose between option A and option B. By choosing option A, you receive a voucher, whereas by choosing option B, you receive an additional payment. After you have reached all of your decisions, one of the 21 scenarios will be chosen randomly for you and your assigned partner. Thereafter, the consequences of the chosen decision are realized, i.e., in the case that you chose option A under this scenario, you will be given the voucher and in the case that you chose option B, you will receive the respective amount from the selected scenario. The same applies to your partner. On the next page, you will receive details about the payment procedure.

E.9.2 Further Procedure

After you confirmed your decision on the decision screen, a screenshot will be taken from this decision screen. From the decision screen of your partner, a screenshot will be taken in the same way. At the end of today's experiment, you will get the screenshot with the decision of your partner displayed, and your partner will get the screenshot with your decision. You will not receive any further information about your partner, and your partner will not receive any further information about you.

Data protection

The subsequent analysis of all data is carried out anonymously so that your decisions can never be linked to your person. Your anonymity is therefore always guaranteed, and the information about your decisions is only used for anonymized data analysis.

Please note:

This is not a thought experiment: All information given in these instructions is true. In particular, all actions are performed exactly as they are described. This fundamentally applies to all studies of the Bonn Laboratory for Experimental Economic Research, as well as to this study.

If you still have separate questions, you may send them to experiment@briq-institute.org after the study.

E.9.3 Comprehension Questions

Question 1: For this question, hypothetically assume that you made your decisions as follows: [Example of MPL decision with switching point at 16€Also assume that the decision situation number 14 (highlighted in gray) [number 14 is 13€] was randomly selected for the real payout for you and your partner.
Question 1a: How many euros would you receive as an additional payment (not in the form of the voucher) in this case?
Question 1b: Would you have received the voucher?
Question 1c: If, in this decision situation, you had chosen option B instead of option A, how many euros would you have received as an additional payment?
Question 1d: Would you have received the voucher in this case?
Question 2: For this question, hypothetically assume that you made your decisions as follows: [Example of MPL decision with switching point at 8€] Also assume that the decision situation number 10 (highlighted in gray) [number 10 is 9€ was randomly selected for the real payout for you and your partner. How many euros would you receive as an additional payment in this case?
Question 2a: How many euros would you receive as an additional payment (not in the form of the voucher) in this case?
Question 2b: Would you have received the voucher?
Question 2c: If, in this decision situation, you had chosen option A instead of option B, how many euros would you have received as an additional payment?
Question 2d: Would you have received the voucher in this case?
Question 3: For this question, hypothetically assume that you made your decisions as shown in the table on the left, while your partner made their decisions as shown in the table on the right. [Example of MPL decision with switching point at 18€for the left, and 20 for the right table] Also assume that the decision situation number 3 (highlighted in gray) [number 3 is 2€] was randomly selected for the real payout for you and your partner. What are the consequences for you and your partner?